Mathematics, referred to as Gaṇita is an integral part of Indians from very ancient times[1]. Ancient Indians developed several concepts of mathematics primarily in the course of solving several real-life problems that they were facing. For example, one of the Vedāṅgas is Jyotiṣa, which confines itself with required mathematical concepts, measurements, and approximation techniques to make certain predictions of the movement of celestial bodies in the sky. Similarly, another Vedāṅga, Kalpa required principles of geometry to construct Vedic altars of many different shapes. Buddhist and Jain were also deeply interested in mathematical concepts and produced canonical works discussing some useful mathematical concepts.

What started as a requirement at an early stage continued through the post-Vedic period as an uninterrupted tradition and many useful contributions to the field of mathematics were made throughout Indian history. Over the years, the contributions spanned almost all modern areas of mathematics. These include algebra, geometry, trigonometry, indeterminate equations, square, cube, and their roots, permutation and combinations, numerical methods and approximations, infinite series, to name a few.

8.1 UNIQUE ASPECTS OF INDIAN MATHEMATICS

Indian mathematics is unique, and it vastly differs from the modern approaches. A few of them are worthy of mention:

(a) There is a popular thinking among many that the world is divided into those who know and love mathematics and those who don't. This separation was primarily because we are educated that mathematics works with left-brain functionalities and literature works with right-brain functionalities. Therefore, the design of the pedagogy and delivery have kept these away in two separate compartments. In contrast, Indian mathematics is a seamless blend of poetry, literature, logic, and mathematical thinking weaved into a single work. All great works of mathematics are invariably great literary works too and they appeal to everyone on account of this natural blend. Therefore, there is no fear or stress in learning Mathematics.

> - Indian Mathematics is a seamless blend of poetry, literature, logic, and mathematical thinking weaved into a single work.
> - The use of sūtras and pithy verses is characteristic in the Indian Mathematical tradition to convey complex ideas and concepts.

(b) Mathematics was considered as a part of life. This is why Mathematics could be found in temple inscriptions, literary work addressing issues of life, and in a discussion on religion or spirituality. Bhāskarācārya in his Līlāvatī, for example, brings interesting mathematical concepts by posing interesting riddles to a student and solving them. Similarly, Śaṇkarācārya in his bhāṣya on Yoga-sūtra refers to the decimal place value system as an example while discussing a philosophical issue.

(c) The use of sūtras is characteristic in the Indian tradition to convey ideas and concepts. These improve the retention of complex ideas and details very easily. Indian mathematics uses these mechanisms as much as possible. For example, there is a simple sūtra for remembering the pattern that constitutes a binary cycle of length three based on eight groups defined by Piṅgala in his Chandaḥ-śāstra. Āryabhaṭa's uniqueness lies in his ability to use sûtras to bring the utmost simplicity to describe complex information. A shining example is his description of the entire sine table using a verse of just two lines (see the last Endnote in Chapter 1). One who is familiar with Āryabhaṭa's system of numeration will be able to remember the sine table with very little effort.

(d) There has been an uninterrupted tradition of mathematical thinking and it has widely spread across the length and breadth of India. Mathematical concepts were developed by those from Gāndhāra (modern-day Afghanistan) to those in Bengal, as well as by those from Kashmir to Kerala.

(e) One of the important characteristics of Indian Mathematics is its algorithmic approach and the fact that it allows for approximate solutions based on the needs of real-life situations. Indian mathematicians also adopted what is today referred to as the constructive approach where the emphasis is on finding a procedure to solve a problem rather than merely seeking proofs of existence of a solution.

8.2 GREAT MATHEMATICIANS AND THEIR CONTRIBUTIONS

Before we discuss some key mathematical concepts developed by Indians, it will be interesting to have a bird's eye view of some of the great mathematicians and their contributions. The number of mathematicians India has produced, and their contributions are significant. Table 8.2 illustrates the major contributions. A closer look at the table reveals a continuous stream of mathematical theory building. Several areas of modern mathematics have been addressed. These include for example Fibonacci series, Pell's equation, and Pascal's triangle. In this chapter, we shall take a few examples from different areas of mathematics and briefly sketch the contributions made by Indians in the past and illustrate a few methods. This is to provide a curtain riser view of the Indian mathematical tradition and its contribution to the development of science, engineering, and technology.

- Indian Mathematics dealt with almost all areas of modern mathematics for more than 1000 years.
- Mathematical concepts found in Vedic texts, Buddhist and Jain works suggest that this culture is several thousand years old.

8.3 ARITHMETIC

Arithmetic is the branch of mathematics dealing with computation using numbers. Arithmetic is part of the daily life of any society. Commercial operations and trade require handling numbers and basic mathematical operators such as addition, subtraction, multiplication, and division. Indian arithmetic was quite sophisticated by the time of Āryabhaṭa in the 5th century CE. This is primarily because of a fully developed decimal place value system employing 0 to 9. In the 7th century, Brahmagupta established the use of 0 as a number and gave rules for multiplication, division, addition, and subtraction of zero. Indian decimal place value system was well known by the middle of 7th century as evident from the following observation of the Syriac bishop Severus Sebokht, "Indians possess a method of calculation that no word can praise enough. Their rational system of mathematics, or their method of calculation. I mean the system using nine symbols.[2]"

TABLE 8.2 Ancient Indian Mathematicians and Their Salient Contributions

Sl. No.	*Details of the Work/Mathematician*	*Period, Location*	*Salient Contributions*
1	**Vedic Texts**	3000 BCE or earlier	The earliest recorded mathematical knowledge, number system, Pythagorean type triplets; Decimal system of naming numbers, the concept of infinity;
2	**Lagadha** – Vedāṅga-jyotiṣa	~ 1300 BCE	Astronomical concepts; a mathematical model for sun-moon movement in time; equinoxes and solstices;
3	**Śulba-sūtras** (Baudhāyana, Āpasthamba, Kātyāyana and Mānava Śulba-sūtras)	800–600 BCE	Earliest Texts of Geometry; Approximate value of the square root of 2, and π. Exact procedures for the construction and transformations of squares, rectangles, trapezia, etc.
4	**Pāṇini** – *Aṣṭādhyāyī*	500 BCE; Śalātura (in Khyber province in Pakistan)	Algorithmic approaches; Originator of the Backus–Naur Form (BNF), used in programming languages today, Context-sensitive rules, Arrays, inheritance, polymorphism;
5	**Piṅgala** – *Chandaḥ-śāstra*	300 BCE	Binary sequences; Conversion of Binary to Decimal system and vice versa; 'Meru Prastara' (Pascal's triangle); Optimal Algorithms to calculate powers; Zero as a Symbol;
6	**Bauddha Mathematical Works**	about 500 BCE to 500 CE	Multi-valued logic, Discussion of indeterminate and infinite numbers;
7	**Jaina Mathematical Works** – *Sūrya-Prajñapti, Jambūdvīpa-prajñapti, Bhagavatī-sūtra, Sthānāṅga-sūtra, Uttarādhyayana-sūtra, Tiloyapannati, Anuyoga-dvāra-sūtra*	200 BCE to 300 CE	Concept of logarithms, large numbers; algorithms for raising a number by a power; the arc of a circle; combinatorics; mensuration; Decimal system; Approximation of π;
8	**Āryabhaṭa** – *Āryabhaṭīyam*	476–550 CE; Kusumapura, near Pataliputra, Bihar	Concise verses; Algorithm for square root, cube root, Place value system; Sine table; geometry; quadratic equations; Linear indeterminate equations; Sums of squares and cubes of numbers; Planetary astronomy; Plane and spherical trigonometry;
9	**Varāha Mihira** – *Bṛhat Samhitā, Bṛhat-jātaka, Pañca-siddhāntikā*	482–565 CE, Ujjain, Madhya Pradesh	Summary of five ancient siddhāntas; Sine table, trigonometric identities; ($\sin^2 + \cos^2$); combinatorics; Magic squares;
10	**Bhāskara I** – *Commentary on Āryabhaṭīya, Laghu-bhāskarīyam and Mahā-bhāskarīyam*	600–680 CE; Vallabhi region, Saurashtra, Gujarat	Expanded Āryabhaṭa's work on Integer solution for indeterminate equations; Approximate formula for the sine function, Planetary Astronomy;

(*Contd.*)

TABLE 8.2 Ancient Indian Mathematicians and Their Salient Contributions (*Contd.*)

Sl. No.	*Details of the Work/Mathematician*	*Period, Location*	*Salient Contributions*
11	**Brahmagupta** – *Brahmasphuṭa-siddhānta; Khaṇḍakhādyaka*	598–668 CE; Bhillamala in Rajasthan	Rules of arithmetic operations with zero and negative numbers, Algebra (Bījagaṇita); linear and quadratic indeterminate equations; Pythagorean triplets, Formula for the diagonals and area of a cyclic quadrilateral; notion of arithmetic mean.
12	**Virahāṅka** – *Vṛttajātisamuccaya (in Prākṛt)*	~600 CE	Fibonacci numbers; Moric metres.
13	**Śrīdharācārya** – *Triśatikā and Pāṭīgaṇita*	870–930 CE; Bhūrisṛṣṭi or Bhurshut village, Hugli, West Bengal	Arithmetic, Algebra, and Commercial Mathematics; Approximation of square root of a non-square number; Quadratic equations; Practical applications of algebra;
14	**Mahāvīrācārya** – *Gaṇita-sāra-saṅgraha*	800–870 CE; Gulbarga Karnataka;	A comprehensive, exclusive textbook on mathematics covering arithmetic-geometry- algebra. Continuing the ancient Jaina mathematics tradition; permutations and combinations; arithmetic and geometric series; the sum of squares and cubes of numbers in arithmetic progression;
15	**Jayadeva**	10th Century CE or earlier	Cakravāla (cyclic) method for solution of the second-order indeterminate equation.
16	**Shrīpati** – *Gaṇita-tilaka, Siddhānta-śekhara, Dhikoṭidākaraṇa, etc.*	1019–1066 CE; Rohiṇīkhaṇḍa, Maharashtra	Planetary Astronomy
17	**Bhāskarācārya (Bhaskara-II)** – *Līlāvatī on arithmetic and geometry; Bījagaṇita on algebra; Siddhānta-śiromaṇi on astronomy; Vāsanābhāṣya on Siddhānta-śiromaṇi.*	1114–1185 CE; Hailed from Bijjadavīda	Canonical textbooks used all over India, Detailed explanations including Upapatti (demonstration or proof); addition formula for sine function. Surds; permutations, and combinations; Solution of indeterminate equations, Ideas of calculus, including mean value theorem, planetary astronomy; construction of several instruments;
18	**Nārāyaṇa Paṇḍita** – *Gaṇita-kaumudī – a treatise on arithmetic and Bījaganita-aātāmśa – a treatise on algebra.*	1325–1400 CE;	Advanced textbooks taking forward the works of Bhāskarācārya, further properties of cyclical quadrilaterals, summation and repeated summations of arithmetic series, theory and construction of Magic squares, further developments in combinatorics.

Sl. No.	Details of the Work/Mathematician	Period, Location	Salient Contributions
19	**Mādhava of Saṅgamagrāma**	1340–1425 CE; Sangama Grama, in Kerala	Founder of Kerala School of Mathematics – a pioneer in the development of calculus; Infinite series and approximations for π, Infinite series and approximations for cosine and sine functions
20	**Parameśvara,** – *Dṛgganita, Siddhāntadīpikā;* Commentaries on Āryabhaṭīyam, Mahā-bhāskarīya; Laghu-bhāskarīya, Līlāvatī, and Sūryasiddhānta	1360–1460 CE; Alathiyur, (near Tirur), Kerala	Properties of Cyclic quadrilateral; iterative techniques.
21	**Nīlakaṇṭha Somayājī,** *Tantra-saṅgraha; Āryabhaṭīya-bhāṣya, Siddhānta-darpaṇa*	1444–1544 CE; Near Tirur, Kerala	Irrationality of π, basic ideas of calculus; revised planetary theory, which is a close approximation to Kepler's model; Exact results in spherical astronomy
22	**Jyeṣṭhadeva** – *Yukti-bhāṣā*	1500–1575 CE; Kerala	Hailed as the first textbook of Calculus; detailed explanations and proofs of the infinite series given by Mādhava
23	**Śaṅkaravāriyar** – *Kriyākramakarī commentary on Līlāvātī and commentary on Tantra-saṅgraha*	1500–1569 CE Kerala	Explanations and Proofs of the results and procedures given in Līlāvatī
24	**Gaṇeśa Daivajña** – *Buddhi-vilāsinī (commentary on Līlāvatī);*	1504 CE; Nandi Grama, Nadod, Gujarat	Explanations and Proofs of the results and procedures given in Līlāvatī
25	**Kṛṣṇa Daivajña** – *Bījapallva- Commentary on Bījagaṇita of Bhāskarācārya*	1600 CE Delhi	Explanations and Proofs of results and procedures given in Bījagaṇita
26	**Munīśvara** – *Siddhānta-sārvabhauma, commentary on Līlāvatī; Pātisāra;*	17th Century CE; Varanasi	Explanations and Proofs of the results and procedures given in Līlāvatī; trigonometric identities
27	**Kamalākara** – *Siddhānta-tattva-viveka*	1616–1700 CE; Varanasi, Uttara Pradesh	Addition and subtraction theorems for the sine and the cosine; Sines and cosines of double, triple, etc., angles.

8.3.1 Square of a Number

Āryabhaṭa identifies the geometric aspect and the mathematical operation of the square in his definition. As per his definition[3] varga is a square and is also a geometric object whose sides are equal. Bhāskara–I (629 CE) in his commentary on Āryabhaṭīya provides an algorithm for finding the square of any number. The verse that explains the algorithm is given below:

- An algorithm for computation of cube root was given for the first time by Āryabhaṭa.
- Brāhma-sphuṭa-Siddhānta, gives a good description of calculations with positive and negative numbers, Zero, fractions and surds.

अन्त्यपदस्य वर्गं कृत्वा द्विगुणं तदेव चान्त्यपदम् ।
शेषपदैराहन्यात् उत्सार्योत्सार्य वर्गविधौ ॥

antyapadasya vargaṃ kṛtvā dviguṇaṃ tadeva cāntyapadam |
śeṣapadairāhanyāt utsāryotsārya vargavidhau ||

8.4 GEOMETRY

As we saw in Chapter 2, Yajñas formed a very important part of Vedic life. The performance of Vedic rituals involved the construction of a variety of Vedic altars (*yajña-vedīs*) as per certain specifications. The Brāhmaṇa portion of the Yajurveda contains details about the arrangement of the sacrificial ground and the construction of the altars. For this purpose, one first fixes cardinal directions and then goes on to construct altars of different shapes and dimensions using prefabricated bricks. Śulba-sūtras give exact methods for the construction of such altars. On account of these, the birth of Indian geometry can clearly be traced to the Vedic times.

Śulba-sūtras are a part of the Vedāṅga. These sūtras give information on the methods of the layout of the vedi, citi, and the maṇḍapa. Śulba means a thread and with the help of a thread and a pin (or a pole), construction methods for various shapes have been described in the śulba-sūtras. The four śulba-sūtras attributed to Baudhāyana, Mānava, Āpastambha, and Kātyāyana are the predominant ones. We also have śulba-sūtras due to Maitrāyaṇa, Varāha, and Vādhūla. The oldest of the śulba-sūtras attributed to Bodhāyana is estimated to be written prior to 800 BCE. Śulba-sūtras are manuals for the construction of Vedic altars. Therefore, Śulba-sūtras also discuss the manufacturing of bricks, thereby indicating a good knowledge of the materials and manufacturing processes which seems to have prevailed in those times.

8.4.1 Property of Right-angled Triangle in Śulba-sūtras

In current mathematical texts, we are taught an important theorem being attributed to Pythagoras (570–495 BCE). However, the first general statements of this so-called Pythagoras theorem is actually found in the Śulba-sūtras. In the Baudhāyana-śulba-sūtra (prior to 800 BCE), the theorem is stated in the following form (this result is known by name *Bhuja-koti-karṇa-nyāya)*: "The sum of the areas of the squares formed by the length and breadth of a rectangle equals the area produced by the diagonal of the rectangle"[7]. Let us consider a rectangle of length a and breadth b. Let the diagonal of the rectangle be c. According to the Baudhāyana formula, $a^2 + b^2 = c^2$. This is graphically illustrated in Figure 8.2.

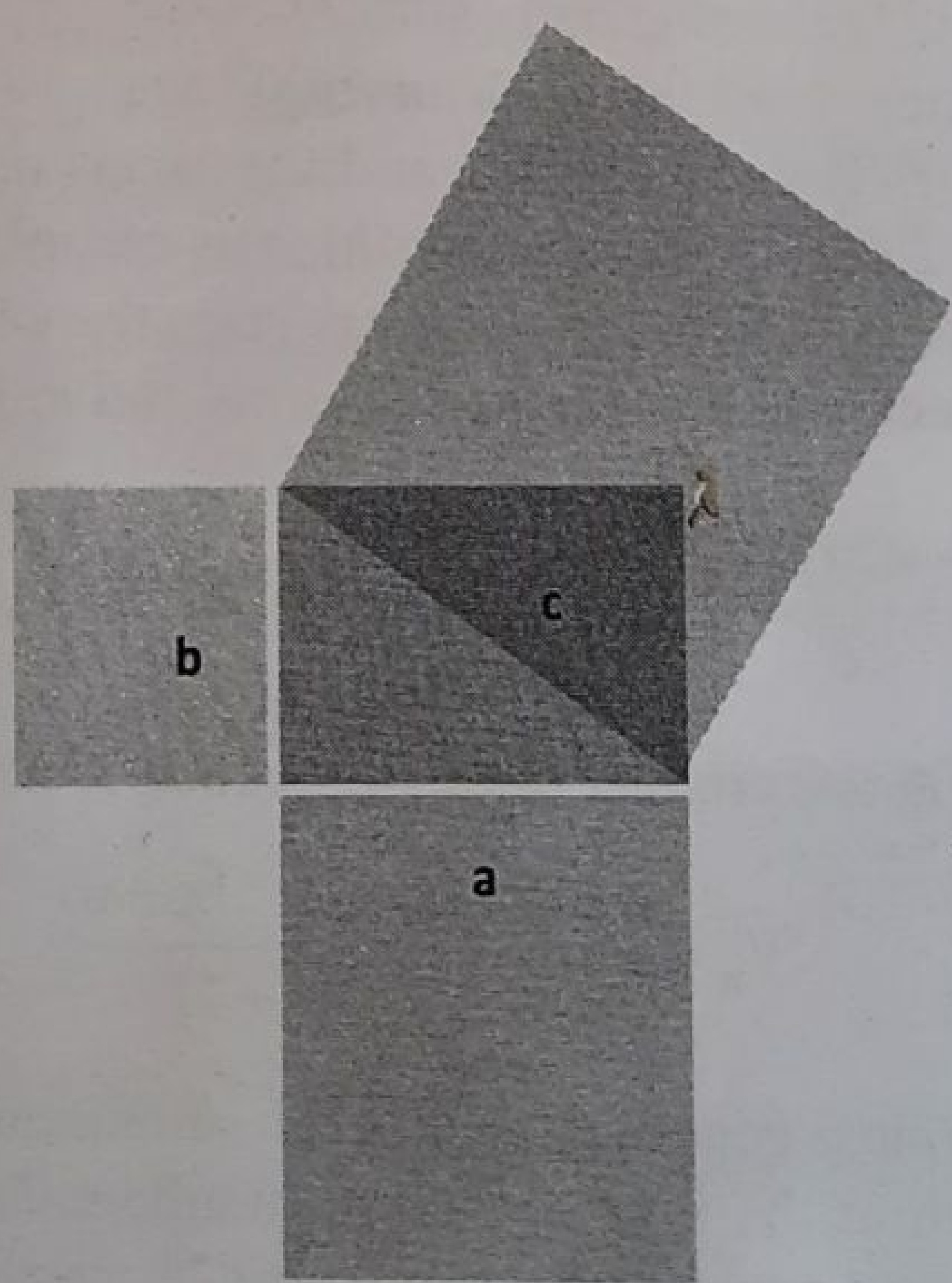

FIGURE 8.2 Baudhāyana Formula for Right-angled Triangle

discussed in Chapter 6 of the book, we can compute π as follows:

$$\pi = \frac{2827433388233}{9*10^{11}} = 3.141592653592\ldots$$

The following Table 8.3 summarises the different approximations to value of π that were discovered in the Indian tradition right from the Vedic times to the time of Ramanujan.

TABLE 8.3 History of Approximations to π by Indian Mathematicians

	Value of π	*Accuracy (Decimal places)*	*Method*
Śulba-sūtras (around 800 BCE)	3.08888	1	Geometrical
Jaina texts (500 BCE)	$\sqrt{10}$ =3.1623	1	Geometrical
Āryabhaṭa (499 CE)	$\frac{62832}{20000} = 3.1416$	4	Polygon doubling ($4.2^8 = 1024$ sides)
Bhāskarācārya (Līlāvatī)	3927/1250 = 3.1416	4	Polygon doubling
Mādhava (1375 CE)	$\frac{2827433388233}{9*10^{11}} = 3.141592653592\ldots$	11	Infinite series with end corrections
Ramanujan (1914 CE)		17 million	Modular equation

8.5 TRIGONOMETRY

Trigonometry is called *jyotpatti,* the science of computation of chords in Indian mathematics. Consider a circle of radius R as shown in Figure 8.5. DA = $R\theta$, is an arc. AB = $R \sin\theta$, is called the jyā corresponding to the arc, $R\theta$. Earlier it was called jyārdha or 'half a bow-string', but later it was just called 'jyā'. R is taken to be $\frac{21600}{2\pi} \approx 3438$, where the circumference of the circle is 21,600 units (number of 'minutes' in a radian). OB = $R\cos\theta$ is called the koṭijyā or kojyā or cojyā. Jīvā was another name for jyā. When this was transmitted to Arab countries,

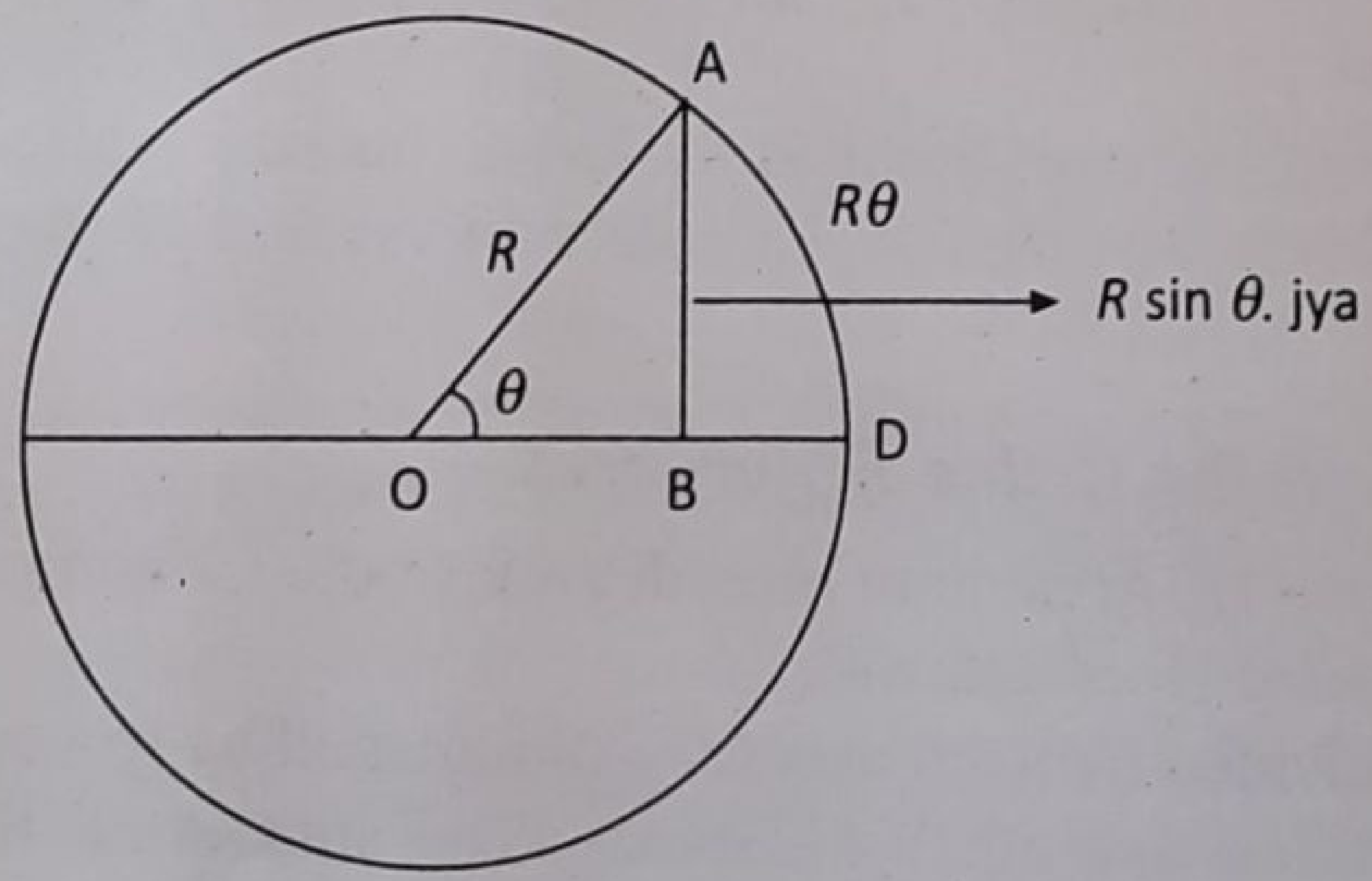

FIGURE 8.5 The Jyā and Cojyā in Indian Mathematics

Let $R_1, R_2, R_3, \ldots, R_{24}$, denote the 24 R sines and δ_1 $(=R_1)$, $\delta_2, \delta_3, \delta_4, \ldots, \delta_{24}$ denote the 24 R sine-differences. Then, according to the above verse,

$$\delta_2 = R_1 - \frac{R_1}{R_1}$$

$$\delta_{n+1} = \delta_n - \frac{\delta_1 + \delta_2 + \delta_3 + \cdots + \delta_n}{R_1} = \delta_n - \frac{R_n}{R_1}$$

Nīlakaṇṭha's Formula for R sine Differences

Nīlakaṇṭha (1500 CE) provided a much accurate estimate of R sine differences by providing a correction to the above formula. According to him, "the first R sine divided by itself and then diminished by the quotient gives the second R sine difference. To obtain the other R sine differences, divide the preceding R sine by the first R sine and multiply the quotient by the difference between the first and second R sine differences and subtract the resulting product form the preceding R sine difference". It is stated mathematically below:

$$\delta_{n+1} = \delta_n - \left(\frac{R_n}{R_1}\right)(\delta_1 - \delta_2)$$

This is the same as the relation

$$2 \sin nx - \sin\{(n+1)x\} - \sin\{(n-1)x\} = (\sin nx/\sin x)\ (2 \sin x - \sin 2x)$$

This approach to computing R sine difference was extraordinary to be conceived at the time of Āryabhaṭa. French mathematician and astronomer Jean Baptiste Joseph Delambre, (1749–1822) paid rich tributes to Āryabhaṭa's analytical method, "the method is curious: it indicates a method of calculating the table of sines using their second differences …, the differential process has not up to now been employed except for Briggs, who himself did not know that the constant factor was the square of the chord. Here then is a method, the Indians possessed, and which is found neither among Greeks not amongst the Arabs[10]".

8.6 ALGEBRA

Algebra became the core of mathematics in the 20th century, which many historians termed as 'algebraization of mathematics'. Algebra provided simplicity, clarity, and precision, to the mathematicians. Algebra is one of the main areas of contribution by ancient Indians as they considered it as a subject of great utility. The ancient Indian name for the science of algebra is bīja-gaṇita. Bīja means 'element' or 'analysis' and gaṇita 'the science of calculation.'

The science of algebra is broadly divided by the Indian mathematicians into two major parts. Of these, one deals with analysis (bīja). The other part discusses concepts that are essential for analysis. It includes the laws of signs, the arithmetic of zero (and infinity), operations with unknowns, surds, and the linear indeterminate equation (known as kuṭṭaka or pulveriser). Therefore, one is able to identify several important features of algebra in the Indian mathematical works. For instance, symbols were used for unknowns by Brahmagupta (it is also indicated in Āryabhaṭīya and its commentary by Bhāskara). Operations with negative numbers were also introduced by Brahmagupta. Furthermore, we can notice that linear and quadratic equations were solved by Āryabhaṭa and Brahmagupta. The Indian mathematicians recognised few fundamental operations in algebra, viz, addition, subtraction, multiplication, division, squaring and the extraction of the square-root.

Recognition of negative numbers and their treatment in mathematical operations were evident very early. Brahmagupta (628 CE) says: "The sum of two positive numbers is positive, of two negative numbers is negative; of a positive and a negative number is their difference." The use of symbols to denote unknowns, working on equations to solve unknown were the key contributions of ancient Indians. Indians discovered the usage of fundamental arithmetic operators *'yu'* for *yuta* – addition; *'kṣa'* for *kṣaya* – subtraction; *'gu'* for *guṇa* – multiplication; *'bhā'* for *bhāga* – division. They used *'mū'* or *'ka'* for *mūla* or *karaṇi* denoting root. They used the first letter of the names of different colors to denote different unknown variables. With these, Indians analysed and classified equations – called *samīkaraṇa*.

Āryabhaṭa in his work *Āryabhaṭīyam (Gaṇitapāda,* Verse 24), discusses an interesting case of how to determine the two numbers whose product and difference are known[11]. The procedure explained in the verse can be algebraically stated as follows. Let us consider two numbers x and y. Let $x - y = a$ and $xy = b$. Then as per the verse, x and y can be determined as follows:

$$x=\frac{\sqrt{4b+a^2}+a}{2};\ y=\frac{\sqrt{4b+a^2}-a}{2}$$

8.7 BINARY MATHEMATICS AND COMBINATORIAL PROBLEMS IN CHANDAḤ-ŚĀSTRA OF PIṄGALA (300 BCE)

We saw the work of Piṅgala known as Chandaḥ-śāstra in Chapter 2 and Chapter 6. In this section, we shall study his contributions in the field of binary mathematics and combinatorial problems. The metres in Sanskrit (chandas) were analysed by means of two kinds of syllables, laghu (L), and guru (G) from a prosody perspective. Any syllabic metre (varṇa-vṛtta) is there for characterized as a sequence of L and G. If we replace laghu (L) with '1' and guru (G) with '0', then it transforms the metrical pattern into a binary sequence. A binary sequence is a sequence composed of '1's or '0's. For example, 1,0,0,1 and 0,1,0,0 or examples of binary sequences of length 4. In Chapter 8 of Chandaḥ-śāstra, Piṅgala has analysed a host of combinatorial problems relating to metrical patterns or, equivalently, binary sequences. These are as follows:

- *Prastāra:* A procedure by which all possible metrical patterns or, equivalently, binary sequences of a given length are generated sequentially as an array (prastāra).
- *Saṃkhyā:* The process of finding the total number of binary sequences (rows) in the prastāra (array).
- *Naṣṭa:* For a given row number in an array, the process of identifying the corresponding binary sequence directly.
- *Uddiṣṭa:* Given a binary sequence, the process of identifying the corresponding row number in the array (prastāra), directly.
- *Lagakriyā:* The process of finding the number of binary sequences in the array with a given number of '1's or '0's.
- *Adhvayoga:* The process of finding the space occupied by the array (prastāra) (to determine the floor area needed).

These concepts introduced by Piṅgala around 300 BCE are perhaps the earliest instance of computations involving binary numbers. The basic building block of this binary analysis is the

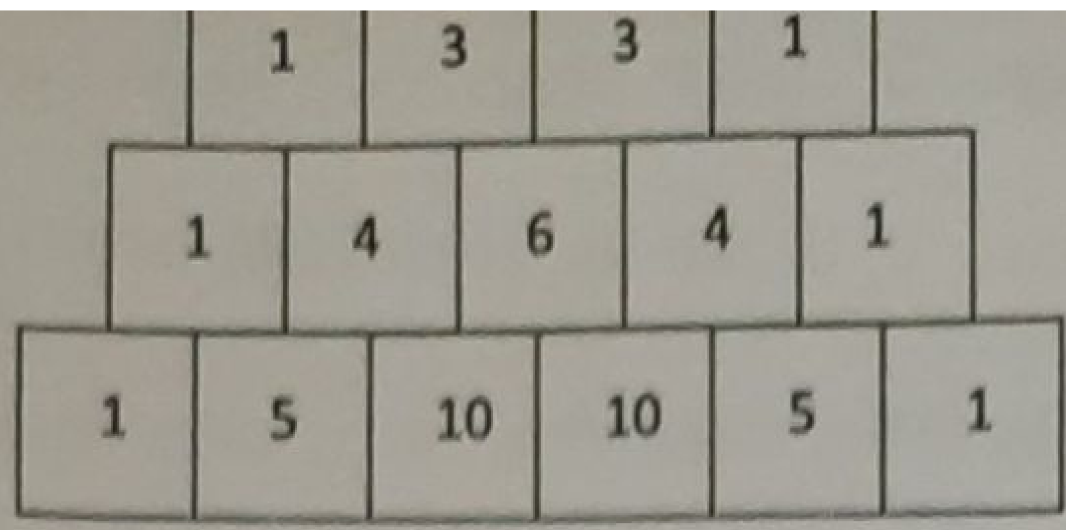

FIGURE 8.8 Varṇa Meru (due to Piṅgala)

Thus, we can see that Piṅgala laid a solid foundation for combinatorial theory and binary mathematics, which was followed by later-day Mathematicians in the country. *Varāhamihira's Bṛhatsaṃhitā* (550 CE), mentions 1820 different combinations that can be obtained by choosing 4 perfumes from a set of 16 basic perfumes. ($^{16}C_4 = 1820$). Varāhamihira also discusses the construction of a *Meru* – tabular form for arriving at this binomial coefficient.

8.8 MAGIC SQUARES IN INDIA

Mathematics can be utilitarian and at the same time a tool for human curiosity, fun, and a matter for mind-expanding exercise. One such example is the magic square problems in arithmetic. An array of numbers containing an equal number of rows and columns is called a magic square when the summation of the numbers in every column and those in every row and each diagonal happens to be the same. On the other hand, if the summation of the 'other' diagonals also is the same, then such a square is called a pan-diagonal magic square. Figure 8.9 is a simple illustration of the magic squares. In the first square [Figure 8.9(a)] the row sums, column sums, and the sum of the main diagonals are 34. However, the other diagonals do not add up to 34. For example, the shaded diagonal (6, 5, 7, and 8) does not add up to 34. Please note that the matrix structure is to be imagined like a torus (a rolled version like a cylinder). Therefore, after 7, the next number in the diagonal will be 8. On the other hand, in Figure 8.9(b), all the diagonals add up to 34. This is a pan-diagonal magic square.

12	3	**6**	13
14	**5**	4	11
7	16	9	2
1	10	15	**8**

(a) Magic square

10	3	**13**	8
5	**16**	2	11
4	9	7	14
15	6	12	**1**

(b) Pan-diagonal magic square

FIGURE 8.9 Magic and Pan-diagonal Magic Square; Magic Sum = 34

The construction of such magic squares has been known in India from very early times. Indian mathematicians specialized in the construction of a special class of magic squares called *sarvatobhadra* or *pan-diagonal magic squares*. Bhadra-gaṇita is the name for the study of magic squares in the Indian mathematical tradition. Various 3 × 3 magic squares are attributed to the ancient astronomer Garga (100 BCE). The work Kakṣapuṭa of Nāgārjuna (100 CE) describes a method for constructing 4 × 4 magic squares, which gives pan-diagonal squares when the magic sum is even. Magic squares have been used to address certain combinatorial issues involving arithmetic. For example, Varāramihira (587 CE) made use of a 4 × 4 magic square to specify varying proportions of four ingredients that can be mixed to prepare different perfumes. 4 × 4 magic squares are found on the gates of buildings, on the walls where shopkeepers transact their business, and on the covers of calendars used by astrologers even to this day. A 4 × 4 square occurs in a Jaina inscription of the 11th century CE, found in the ancient town of Khajuraho. A similar square dated to 1480 CE was found in the Gwalior fort. The first chapter of the notebooks of Srinivasa Ramanujan is on magic squares.

Construction of a 4 × 4 Pan-diagonal Magic Square

Perhaps the easiest magic square to understand is the one presented by Nāgārjuna in his work Kakṣapuṭa, where he provided a mnemonic and a formula to construct a generic magic square. The mnemonic is as follows: "अर्क इन्दुनिधा नारी तेन लग्न विनासनम्" (*arka indunidhā nārī tena lagna vināsanam*). There are 16 syllables in this mnemonic. Using the katapayādi formula (see Chapter 6 for details), the above mnemonic can be converted into 16 numbers. All vowels are zero in kaṭapyādi. Therefore, in the mnemonic, the first number is zero. Populating the 4 × 4 magic square with this number gives us the first template. See Figure 8.10(a) for details. Let the desired sum of the magic square be $2n$. Then using the formula given in Figure 8.10(b) the individual cells of the magic square can be arrived at. For example, if the desired sum is 100, then n = 50. The magic square for this is available in Figure 8.10(c).

0	1	0	8
0	9	0	2
6	0	3	0
4	0	7	0

(a)

n-3	1	n-6	8
n-7	9	n-4	2
6	n-8	3	n-1
4	n-2	7	n-9

(b)

47	1	44	8
43	9	46	2
6	42	3	49
4	48	7	41

(c)

FIGURE 8.10 Nāgārjuna's Scheme for Generating a 4 × 4 Pan-diagonal Magic Square with an Illustration for n = 50 and human life. Indians from time immemorial developed ... natural curiosity. Driven by a fertile imaginative mind they deeply observed and analyzed the natural phenomena and sought answers rationally in the process of developing an ecosystem of knowledge in allied areas in Mathematics.

Astronomy is a branch of science that studies celestial objects, space, and the physical universe as a whole using concepts mainly from Mathematics. The sky and outer space have been a great source of curiosity for mankind forever. Therefore, Astronomy has been a branch of study right from pre-historic times. All ancient civilizations have made a methodical observation of the night sky and developed their understanding of the celestial phenomena. Intense observation of the patterns in the sky using some instruments designed for this purpose has contributed to a better understanding of astronomy. Ancient Indians have also contributed to the study of astronomy in very significant ways and have laid some foundations for its growth in the days to come. We shall see some of the salient aspects of Indian astronomy[1].

9.1 UNIQUE ASPECTS OF INDIAN ASTRONOMY

In modern science, the study of astronomy, space, and celestial objects are considered different and alien entities vis-à-vis human beings. Space is 'out there' and it is a matter of curiosity to understand what it consists of and how it can be studied using mathematical constructs and models. The celestial bodies have no relationship to the society or the culture, except that they influence climatic conditions, rainfall, etc. The Sun, for instance, is imagined as nothing but an inert matter consisting of material and radiating a large amount of energy. How can an entity that brings life to the entire living beings on the Earth be considered lifeless and inert? In contrast, in several cultures including the Indian, the Sun is considered as a living entity and a deity.

Knowledge of astronomy was widely used by all sections of Indian society, not just by the subject matter experts. Villagers, farmers, arts, and craftsmen and householders would know the meaning of Rāśi (Zodiac Sign), Nakṣatra (Star), and months of the calendar, and certain details about every day (Pañcāṅga). They developed some understanding of how seasons are formed, as seasonal changes affect economic activities such as farming, and the general health of individuals. This is common knowledge, even if the citizens are not educated in modern methods. There are cultural-religious aspects too. In India, it is a common tradition for the newly married couple to be shown the pair of stars known as Vasiṣṭha and Arundhatī (corresponding to the visual binary system in the constellation of 'Ursa Major') as part of the marriage ceremony by the priest or the family elders. The celestial binary is held up as a model for the married life of the couple.

- Knowledge of astronomy was widely used by all sections of Indian society.
- The observations and computations of measures related to celestial bodies such as stars and planets have been consistently accurate and ahead of the times as observed by many.

There are several differences between the current (predominantly Western) approach and that of the Indian and other ancient civilizations to astronomy. These are summarized below:

- The celestial entities are an integral part of all living beings on Earth. There is a strong sense of mutual dependence between the earthly and the celestial entities.

- Astronomy, a study of the celestial entities and phenomena is interconnected with cultural practices and daily life. Many events and planning of activities are done with reference to the celestial entities. 'Kālanirṇaya' (Determination of time) was considered the purpose of astronomy. Therefore, consulting the Pañcāṅga is a daily necessity and often the day begins with this activity.
- Astronomy is not a study of some alien entities but an integral and important aspect of one's life. Several concepts and models were developed by ancient Indians to address this requirement. This partly explains the growth and maturity of astronomical thinking in Indian society right from early times.
- Indians developed a systematic procedure of study, observation, data collection, codification, pattern recognition, and analysis. This led to the development of advanced mathematics including arithmetic, geometry, algebra, trigonometry, and the basics of calculus.

9.2 HISTORICAL DEVELOPMENT OF ASTRONOMY IN INDIA

Why astronomy developed in India

The development of astronomy as a systematic area of study was driven by multiple considerations by civilizations the world over. They looked up to the sky for a few important reasons. First, it helped them develop a calendar to predict seasons. The stars in the sky served as a reliable aid for directional and navigational purposes. Moreover, the sky was always considered as an abode of gods, and for astrological predictions. Above all, observing the sky has been a fascinating experience for many. The Indian civilization was driven by similar considerations:

- The Vedic living required methods for fixing an auspicious time for performing various rituals and activities. Knowledge of the cardinal directions is also required to set the Vedic altars.
- In ancient times, there were two major wealth-generating activities: Agriculture and Trade. Agriculture was the main driver of the economy and wealth creation. Ability to predict the weather conditions and understand the mechanism of season formation was a key requirement to improve agricultural productivity.
- The other economic activity is related to international trade. This required a good understanding of the latitude, longitude, and loxodrome. Moreover, there was a need for a calendar as well.
- Astronomy plays a crucial role in navigation. The basic idea in navigation is to follow rising or setting location of a particular constellation for a certain period and then turn towards another constellation. Therefore, knowledge of astronomy becomes crucial. Indians have been regularly using astronomy for maritime purposes and some studies suggest that details of this have been well documented[2].

All these called for a calculation of time and timekeeping, which resulted in the formation and continuous evolution of the Indian calendar system Pañcāṅga, which is an accurate system of calendar even today. Indian Astronomy developed continuously in time right from the Vedic period. For example, in the Atharvaveda-saṃhitā, we are able to see references to stars, planets, and comets[3]. Parāśara-Tantra consists of planet and comet observations made during the 2nd Millennium BCE[4]. The text provides details of the sidereal periods of Jupiter and Saturn.

10	Kerala School **Mādhava of Saṅgamagrāma**– *Veṇvāroha, Sphuṭacandrāpti* **Parameśvara of Vaṭasseri**– *Dṛggaṇita, Bhaṭadīpika, Siddhānta-dīpikā* **Nīlakaṇṭha Somayājī or Somasutvan of Trikkantiyur**– *Tantra-saṅgraha, Āryabhaṭīya-bhāṣya* **Jyeṣṭhadeva** – *Gaṇita-yuktibhāṣā* Acuyta Pisaroti– *Sphuṭanirṇayatantra* Śankaravarman– *Ṣaḍratnamāla*	14th–19th Century 1340–1425 CE 1360–1455 CE 1444–1550 CE 1500–1610 CE 16th century CE 19th century CE	This sch[illegible] mathematical analysis – Derivation of infinite series for π, sine and cosine functions, much before the subject developed in Europe. A major revision of the traditional planetary theory in 1500 CE. Innovations in astronomical computations; Systematisation of the applications of spherical trigonometry to astronomy; Improved theory of eclipses.
11	**Gaṇeśa Daivajña** – *Grahalāghava*	Born 1507 CE	Simplified procedures for calculation of planetary positions, used for preparing almanacs or Pañcāṅgas even now.
12	**Kamalākara**–*Siddhānta-tattva-viveka*	Born 1616 CE	Elaborate work mostly based on Indian concepts and parameters but incorporates elements of the Greek astronomer Ptolemy's system.
13	**Candraśekhara Sāmanta** – *Siddhānta-darpaṇa*	Born 1835 CE	Important modifications in planetary parameters revised the lunar theory, designed simple instruments, reformed the traditional calendar of Odisha.
14	**Rājā Sawai Jai Singh** – *Yantrarāja-racanā, Zij Muhammad Shahi*	1688–1743 CE	Built famous observatories in several parts of North India.

9.3 THE CELESTIAL COORDINATE SYSTEM

A celestial coordinate system is a mechanism for specifying positions of planets, stars, and other celestial objects relative to physical reference points available to an observer situated on the earth. It helps to locate a celestial luminary using a three-dimensional or a spherical plane. Figure 9.1 provides the rudimentary elements of the celestial coordinate system. The celestial pole is an imaginary large sphere concentric to the earth. All the objects in the sky can be projected on the celestial sphere. The celestial north pole and celestial south pole are analogous to the north pole and south pole of the earth respectively. Similarly, the celestial equator is concentric to the earth's equator. Likewise, all elements of the celestial sphere such as the celestial hemisphere are analogous to the earth's hemisphere. The only difference is that it is projected on to a great circle so that all celestial luminaries can be mapped on to the sphere using some celestial coordinates.

Another useful coordinate system is the ecliptic. **Ecliptic** is an imaginary line depicting the Sun's movement around earth. When an observer sees the Sun's path from Earth's reference frame, it appears to move around the Earth in a path that is tilted with respect to the spin axis at 23.5°. Figure 9.1(a) graphically illustrates the ecliptic. The Ecliptic plane sets the reference plane and is the basis for the ecliptic coordinate system which is used for all further astronomical calculations. A third aspect of the coordinate system is the horizontal coordinate system from the perspective and the position of an observer. The horizontal plane with the Zenith and the Nadir are the two poles of this system. The **zenith** is an imaginary point directly 'above' a particular location (highest point), on the celestial sphere and **nadir** is 180° from zenith (the lowest point). It helps to position the stars and other luminaries in the sky relative to the observer's horizon. As we know by experience, the position of the celestial luminaries varies with time and is useful for the observer to locate and track these celestial entities. Figure 9.1(b) is a simple illustration of this. The vector from an observer to a point of interest (such as a star or any celestial body) is projected perpendicularly onto a reference plane. The angle between the projected vector and a reference vector on the reference plane is called the **azimuth**.

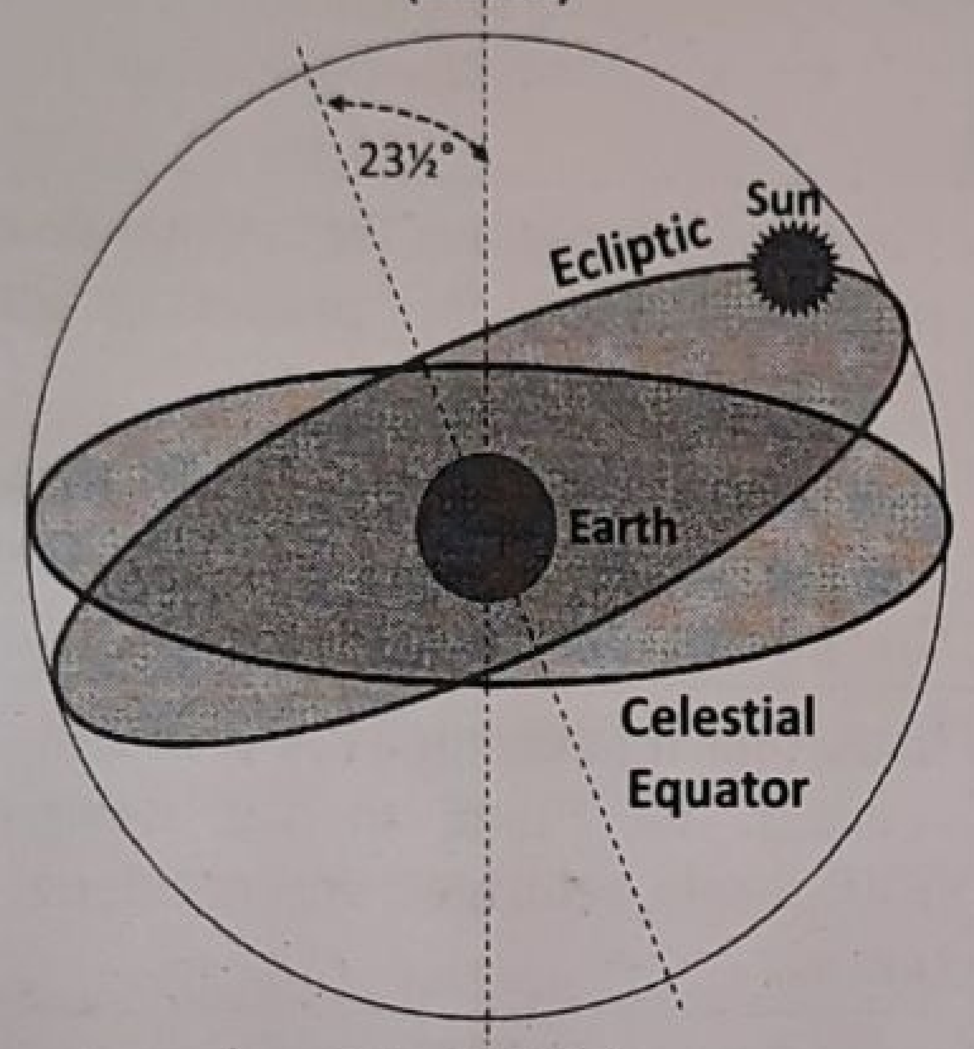

FIGURE 9.1 (a) Illustration of an Ecliptic

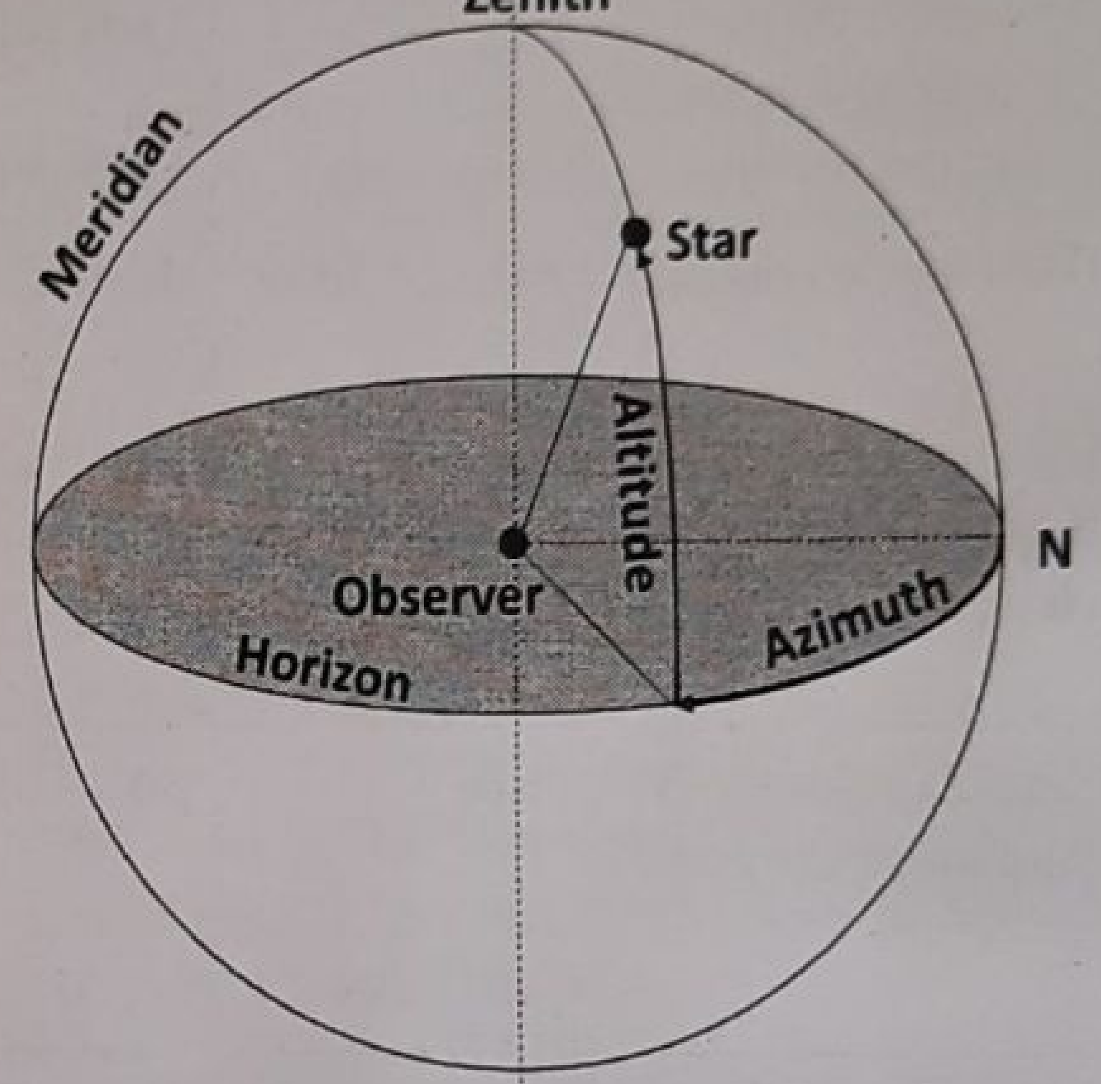

FIGURE 9.1 (b) Azimuth and Zenith

Every celestial object appears to rise in the eastern part of the sky, travel up along a 'diurnal path', and set in the western part. The relative positions of the stars are fixed, and they do not appear to move with respect to each other. However, the Sun, the Moon, and the planets seem to move eastwards in the background of stars (apart from their daily motion from east to west). Let us consider the movement of the Sun, as viewed from the Earth, as shown in Figure 9.2. At the time of Sunrise on a particular day, the Sun is at A which is in the same direction as a star S1. After one day, at the next Sunrise, it would be at B, which is in the same direction as a star S2, east of S1. This change in the position of the celestial objects forms the backbone of computation of various time units such as year, month, and day. The calendaring and other astronomical calculations are done using these basic principles. Further, some of the elements of the seasons are also inferred from analyzing the trajectory of the Sun in the ecliptic.

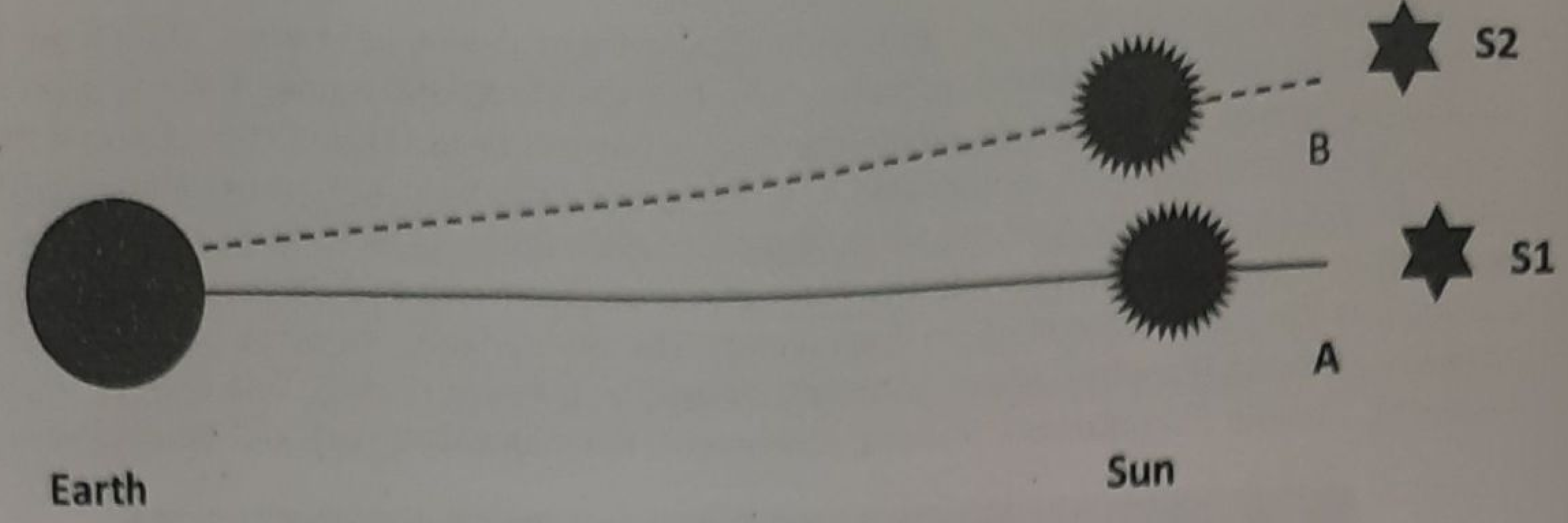

FIGURE 9.2 Motion of the Sun in the Background of Stars

A solstice occurs when the sun's path crosses the extreme north or south points on the earth's equator. In Figure 9.3, when the Sun is at S4, it is at the southernmost point with respect to the equator. This is known as the *winter solstice*. Similarly, when the Sun is at S2, it is at the northernmost point with respect to the equator. That point is known as the *summer solstice*. At S1 and S3, the ecliptic intersects the equator, and these points are known as the *equinoxes*. When the Sun is at equinox, it is an equinoctial day. Between S4 and S2, the Sun moves northwards (known in the Indian system as Uttarāyaṇa), and between S2 and S4, the Sun moves southwards (known as Dakṣiṇāyana). We find Vedic references to these important astronomical concepts. Taittirīya-saṃhitā (6.5.3) observes[8], 'Thus the Sun moves southwards for six months and northwards for six months'. The equinoctial day (viṣuvat) is mentioned in Aitareya-brāhmaṇa (18.4)[9]. Methods to compute the *n*th equinox is available in Vedāṅga-jyotiṣa[10].

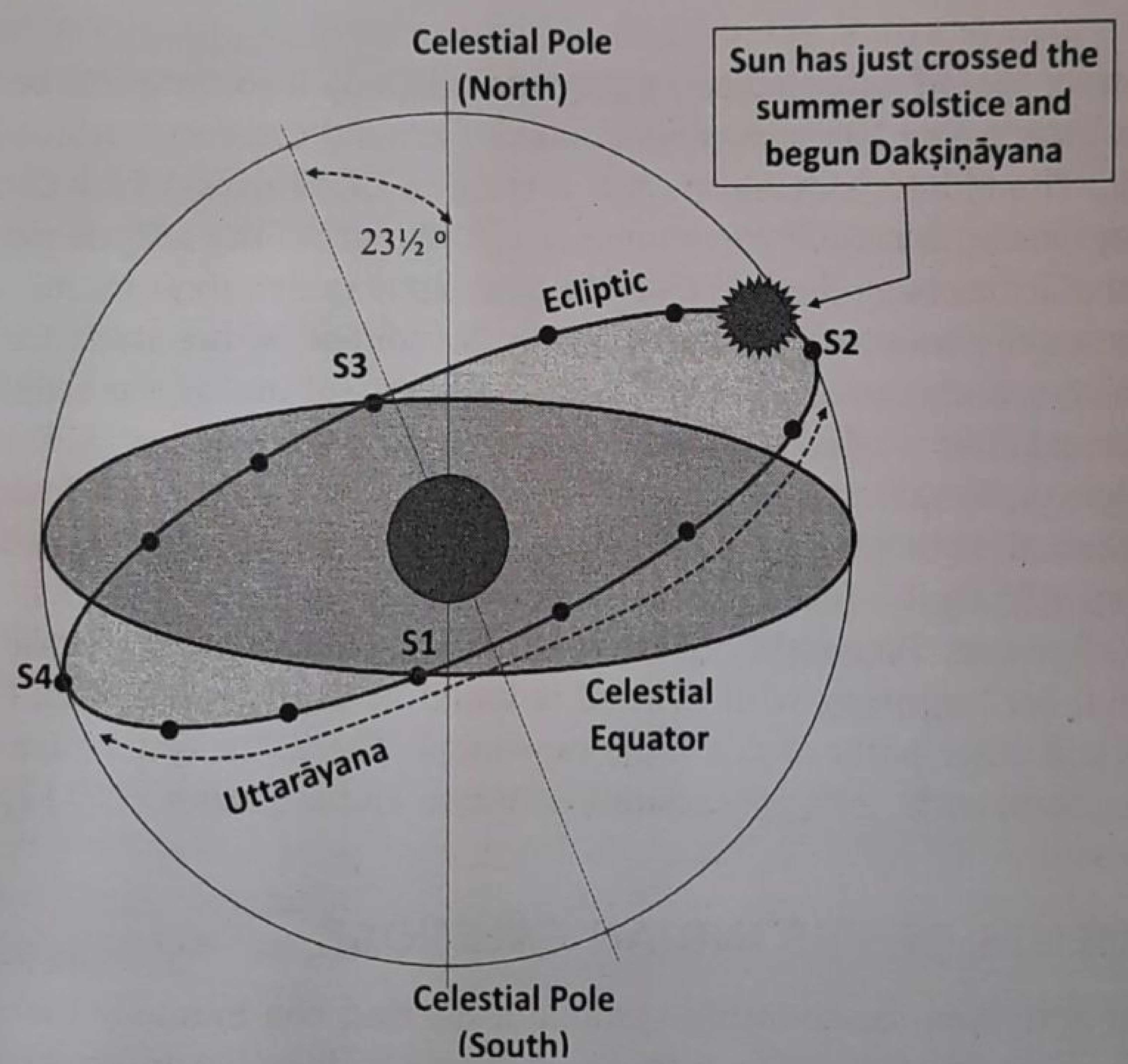

FIGURE 9.3 Illustration of Solstices and Equinoxes

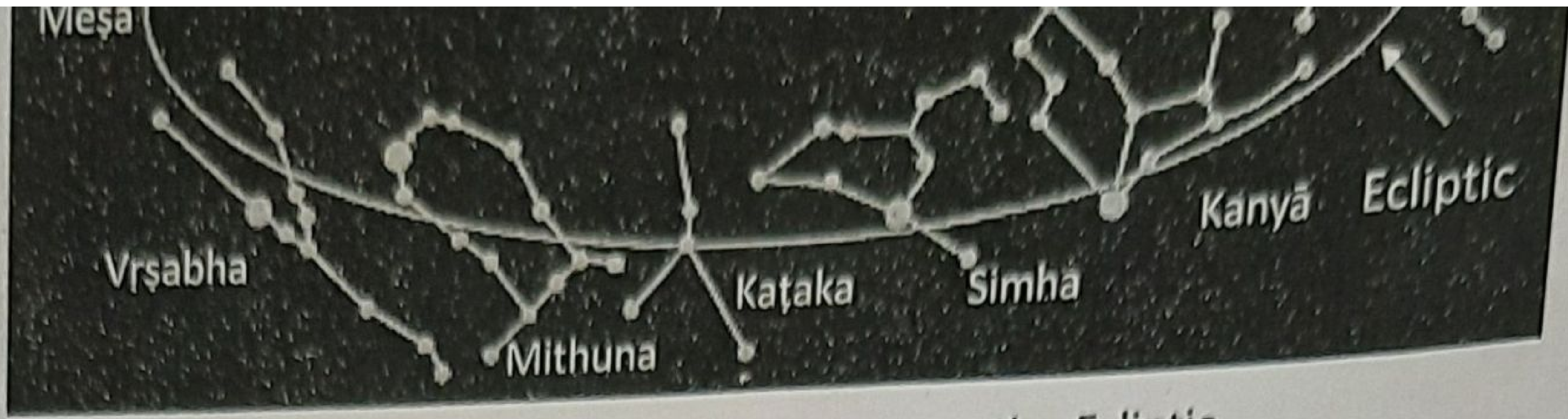

FIGURE 9.4 The Zodiac Signs on the Ecliptic

The 'sidereal period' of an object is the time taken by it to complete one revolution in the background of stars. The Moon moves in an orbit around the Earth, which is slightly inclined to the ecliptic. The Moon's sidereal period is close to 27.32 days, which constitutes one lunar cycle. In other words, the Moon covers nearly 1/27th part of the ecliptic per day. In the Indian system, the zodiac has been divided into 27 equal parts from a fixed initial point in the ecliptic in order to trace the trajectory of the Moon in the context of the stars. Each such division is known as nakṣatra, measuring 13° 20′ or 800 minutes of arc of the ecliptic (360/27). Each division is named after a selected star that is generally prominent or traditionally well-known and is broadly equally spaced in the zodiac. Each day would be associated with a nakṣatra. They are Aśvinī, Bharaṇī, Kṛittikā, Rohiṇī, Mṛgaśiras, Ārdrā, Punarvasu, Puṣya, Āśleṣā, Maghā, Pūrvā Phalgunī, Uttarā Phalgunī, Hasta, Citrā, Svātī, Viśākhā, Anurādhā, Jyeṣṭhā, Mūla, Pūrvāṣāḍhā, Uttarāṣāḍhā, Śravaṇa, Dhaniṣṭhā, Śatabhiṣaj, Pūrvā Bhādrapadā, Uttarā Bhādrapadā, and Revatī. The full list beginning with Kṛittikā is found in Taittirīya-saṃhitā, (4.4.10.1-3), and in Atharvaveda and other parts of the Vedic repository. The 27 nakṣatras are mapped on to the 12 rāśis. Therefore, each rāśi is associated with two and a quarter (27/12) nakṣatra.

9.4 ELEMENTS OF THE INDIAN CALENDAR

A calendar is a method for counting systematically and continuously the successive days by cyclic periods such as year and month. Calendaring is done based on two luminaries in the sky, namely the Sun and the Moon[11]. One of the critical uses of astronomy is to calendar the time and identify seasons, which are related to the relative movement of the Sun with respect to the Earth. The Indian system of astronomy has dealt with this elaborately as evidenced in the Vedic texts. While in the Vedic texts we merely find the description of these elements, the computation details of these are found in Vedāṅga Jyotiṣa, though approximate and simple. Later day mathematicians introduced formal methods for mathematical computations of the calendar. There are three clear time-markers as even casual observations of the sky reveal, viz., day, month, and year. All major civilizations grappled with the problem of how these time units are related to each other.

The Notion of a Year

The elapsed time for the Sun to return to the same star, which is nothing but the period of the revolution of the Earth around the Sun, is the solar year. Solar calendars are based on this. On the other hand, the period of the return of the Moon in opposition (Full moon) or conjunction to the Sun (New moon) in relation to the Earth, is a lunar month. 12 such successive months form the lunar year, and this is the basis of all lunar calendars. Both the calendars are in use today. The solar-year calendar is followed for indicating the dates of the year in the states of Tripura, Assam, Bengal, Odisha, Tamil Nadu, Kerala, and partly Punjab and Haryana while the lunar calendar is followed in other states for the same purpose. But in all states for fixing the dates of religious festivals and for selecting the auspicious time for undertaking many socio-religious activities, the lunar calendar is used. The lunar calendar, however, is pegged to the solar calendar, or more precisely, the lunar months are linked with the solar months to prevent these from getting delinked with the seasons. Therefore, it is more apt to say that in the Indian tradition a luni-solar calendar is followed. Knowledge of the components of the solar and lunar calendar is critical.

What constitutes a year has been variously defined in the Indian system[12]. The Vedic year consisted of 12 months, each of 30 days (known as Sāvana). This results in a year consisting of 360 days. This was synchronized to the seasons by adding 5 days to the calendar. The Ṛgveda (1.164.11) depicts this as, "The wheel (of time), formed with twelve spokes, revolves around the heavens, without wearing out. O Agni, on it, are 720 sons (viz., days and nights)"[13]. According to this verse, a year has 12 months and 360 days. 5 or even 5.25 days were added later. In Yajurveda, there are references to 12 lunar months (known as Vatsara) amounting to 354 days. The synchronization was done using the Ekādaśarātra ceremony to account for 365 days in a year leaving an error of 0.25 days. Subbarayappa and Sarma have compiled the list of astronomical references in various ancient Indian literature pertaining to these aspects[14]. It appears that the authors of the Ṛgveda were aware of the discrepancies between the duration of the Lunar year and Solar year and the need to add an intercalary month for synchronizing the two. Table 9.4 summarises the differences between alternative year cycles in the Indian tradition.

A solar year is approximately 365.2564 days (modern value) and 12 lunar months is approximately 354.3671 days. There is a gap of nearly 10.89 days between the two. Therefore, to align the lunar year with a solar year which is connected with the seasons, an additional lunar month has to be introduced in some years. In Yajurveda (4.4.11) there is a mention of the need to add 2 intercalary months in 5 years[15]. It had been noticed that the Sun and the Moon return together to nearly the same position in the framework of stars, after five years. The concept of Yuga was introduced to synchronize the Solar and Lunar calendars. Two intercalary

TABLE 9.4 Alternative Classification of Year Cycles in the Indian Tradition

	Solar Year	*Sâvana Year*	*Lunar Year*
Year	12 Solar Months (366 days)	12 Sâvana Months (360 days)	12 Lunar Months (354 days)
Month	30.5 days	30 days	29.5 days
Yuga	60 Solar Months	61 Sâvana Months	62 Lunar Months
Remarks	Known as Saṃvatsara	Known as Idâvatsara	Known as Anuvatsara

months Aṁhaspati and Saṁsarpa were added to complete a Yuga.[16] A 5-year yuga cycle is mentioned in Taittirīya and Vājasaneyī saṃhitās.

Solar and Lunar Months

The Sūrya-siddhānta defines the solar month as the time taken by the Sun to traverse a rāśi, which is the difference between the time of ingress of the Sun to one rāśi (saṅkrānti) and to the time of ingress to the next rāśi. As Meṣa is the first rāśi, the time taken by the Sun to traverse this rāśi forms the 1st solar month, and it is called Meṣa in Kerala, and Vaiśākha in Tripura, Bengal, Odisha, Haryana, and Punjab, Bahag in Assam, and Citra in Tamil Nadu. The second solar month is similarly formed by the time taken by the Sun to traverse the second rāśi, which is Vṛṣabha, and so on for other months. The names of the twelve months and six seasons are given in the Taittirīya-saṃhitā (1.4.14; 4.4.11)[17] as "madhu, mādhava (vasanta ṛtu), śukra, śuci (grīṣma ṛtu), nabha, nabhasya (varṣa), īśa, ūrja (śarad ṛtu), sahas, sahasya (hemanta ṛtu), tapas, tapasya (śiśira ṛtu)".

A lunar month (cāndra-māsa) is the time interval between two new Moons (Amāvāsyā), or two full Moons (Pūrṇimā). It consists of two pakṣas; Śukla-pakṣa (bright fortnight) from Amāvāsyā to Pūrṇimā, and the Kṛṣṇa-pakṣa (dark fortnight) from Pūrṇimā to Amāvāsyā. Figure 9.5 illustrates this idea pictorially. A normal lunar year has twelve lunar months. The names of the twelve lunar months are Caitra, Vaiśākha, Jyeṣṭha, Āṣādha, Śrāvaṇa, Bhādrapada,

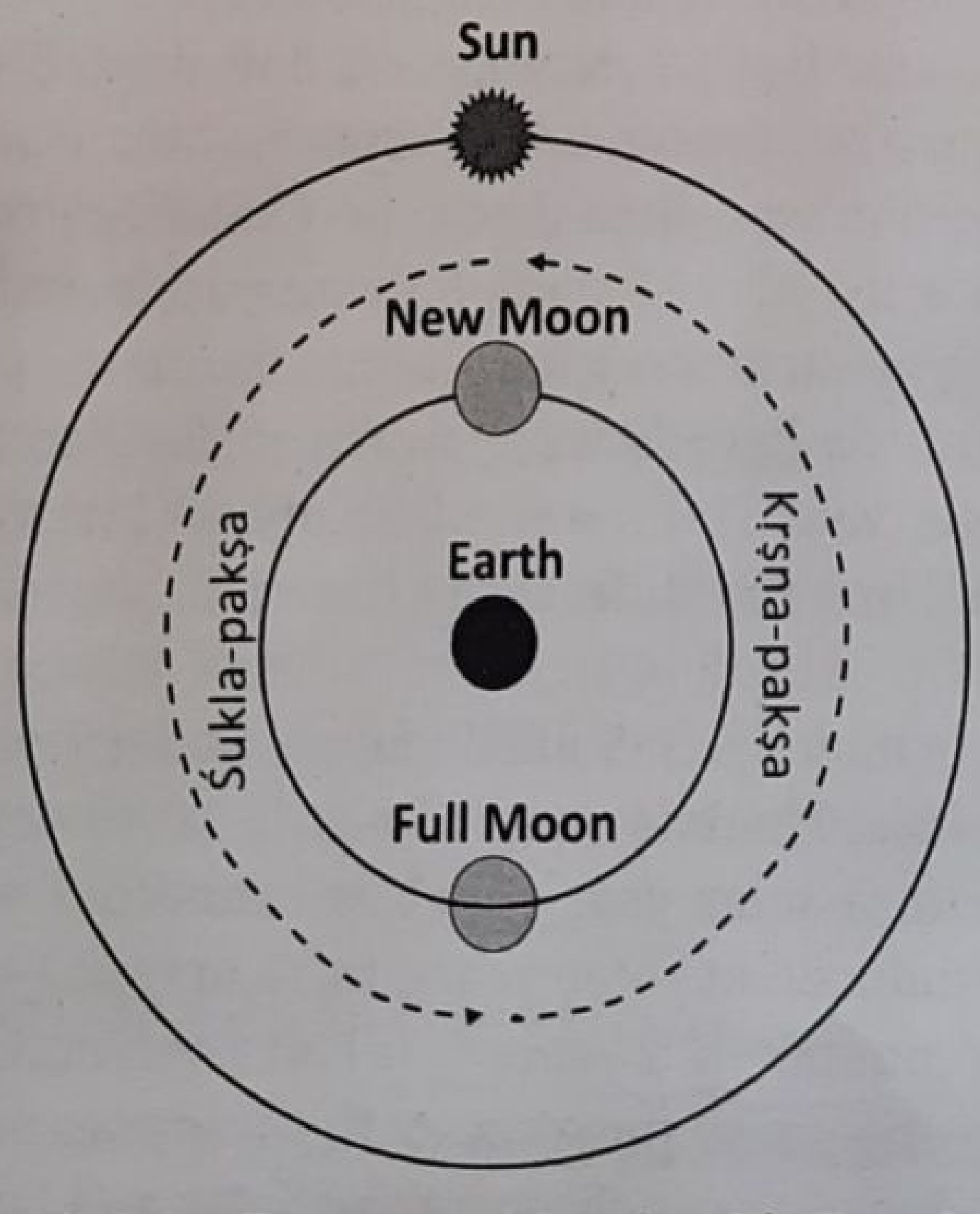

FIGURE 9.5 The Lunar Month with Two Pakṣas

Āśvayuja, Kārtika, Mārgaśira, Puṣya, Māgha, and Phālguna. During most Caitra months the Moon will be close to the star Citrā (Spica), on the full moon day of the month. Similarly, during most Vaiśākha months, the Moon will be near the star Viśākhā on the full moon day in that month. This is the logic behind the nomenclature of the months.

The Notion of Tithi

The Indian calendar is based on the true longitudes of the Sun and the Moon and is related to the actual happenings in the sky. The relative position of the Moon and the Sun on a day determines the tithi. In the Stellar background, the Sun moves at the rate of per day eastwards and the Moon moves at the rate of per day. Tithi captures the angular separation between the Sun and the Moon. It is a time-unit during which the angle between the Sun and the Moon (as viewed from the Earth) increases by 12° or its integral multiples. At Amāvāsyā, the Sun and the Moon are perfectly aligned (that is why we are not able to sight the moon) and the angle is 0°, and the first tithi begins. At the end of the first tithi, the angular separation is 12° and the second tithi begins. The second tithi ends, and the third begins when the angle is 24°, and so on. Figure 9.6 illustrates this pictorially. The 12° is merely an average. In reality, the actual duration of the tithis will vary on account of the perturbing forces affecting the true motions of the Moon and the Sun. The actual duration of a particular tithi may vary from 26 hours 47 minutes to 19 hours 59 minutes.

As we saw, there are rudiments of a calendar with adhikamāsas (intercalary months), and 27 nakṣatras as markers of the Moon's movement. However, the descriptions in the saṃhitās are qualitative. A definite quantitative calendrical system is described in Vedāṅga Jyotiṣa. It is available in Ṛgvedic and Yajurvedic versions. Vedāṅga-jyotiṣa is the first text in India to give mathematical algorithms in astronomy. There is nothing on planetary motion in this work. There are short algorithms for finding tithi, nakṣatra, Sun's position in the sky, etc. The Vedāṅga-jyotiṣa gives rules for calculating some astronomical quantities related to the Sun and the Moon. The motions of the Sun and the Moon (in the background of stars) are assumed to be uniform.

New Moon
Sun = S_0, Moon = M_0; $M_0\hat{E}S_o = 0°$
After 1st tithi
Sun = S_1, Moon = M_1; $M_1\hat{E}S_1 = 12°$
After 2nd tithi
Sun = S_2, Moon = M_2; $M_2\hat{E}S_2 = 24°$
Full Moon
Sun = S_{15}, Moon = M_{15}; $M_{15}\hat{E}S_{15} = 180°$

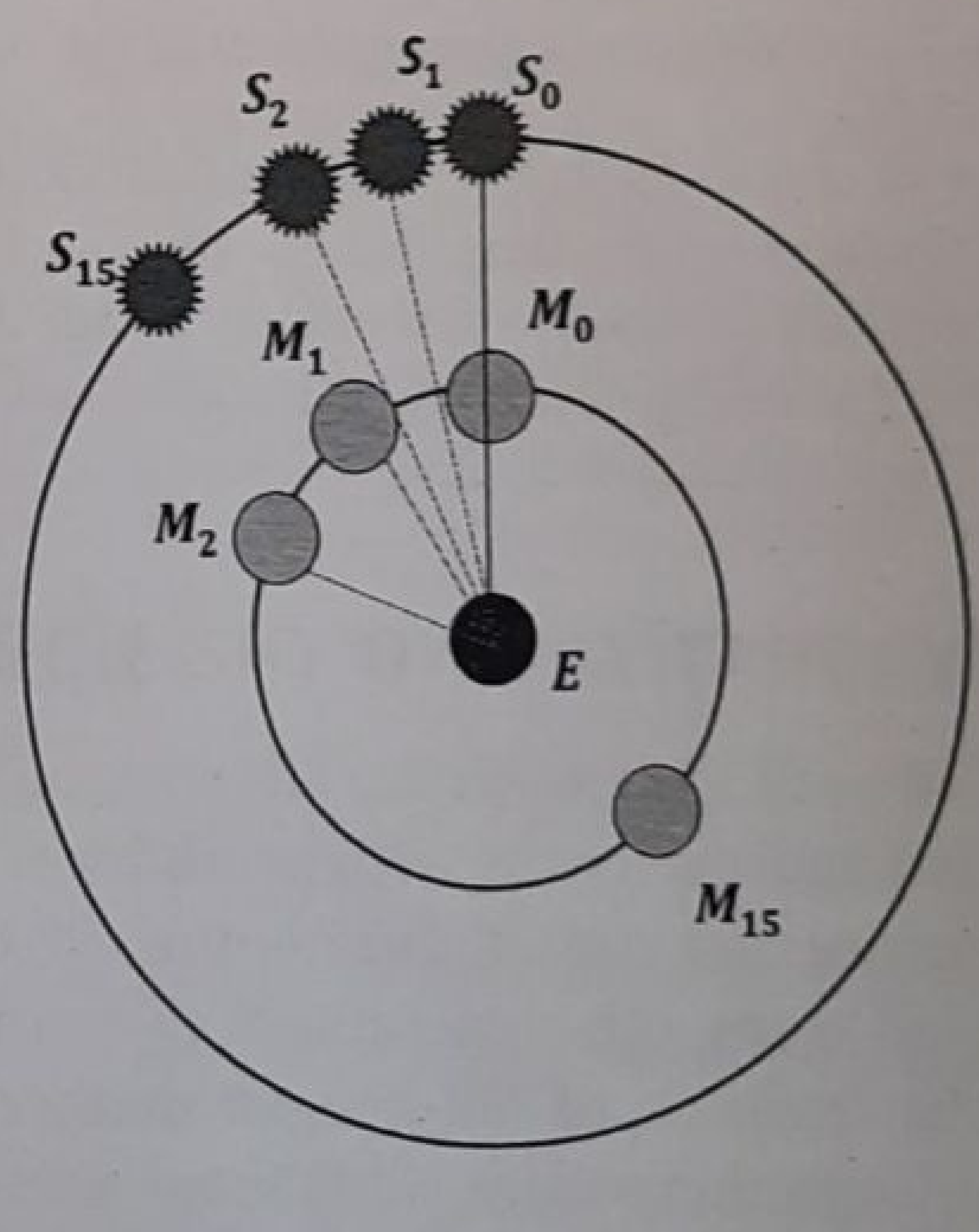

FIGURE 9.6 An Illustration of Tithi

The later astronomical texts of the siddhānta period are far more sophisticated and elaborate, with procedures to calculate planetary positions, eclipses, etc. The calculations are

How to Compute the Quantum of Daytime on any Day?

It is a common observation that sunrise and sunset times are different each day in a year. Hence the quantum of the daytime is not constant. Ancient Indian astronomers had formulated procedures to calculate the daytime on any given day, using their observations. Basically, they had observed that on the equinoctial day, when the sun is on the equator, the durations of daytime, and night-time are equal. On the other hand, the quantum of daytime is the highest at the summer solstice and least at the winter solstice.

Vedāṅga-jyotiṣa, gives a simple arithmetical rule for the duration of the daytime (in verse 22 of the *Ṛgveda* version and verse 40 of *Yajurveda* version):

यदुत्तरस्यायनतो गतं स्याच्छेषं तथा दक्षिणोऽयनस्य ।
तदेकषष्ठ्या द्विगुणं विभक्तं सद्वादशं स्याद्दिवसप्रमाणम् ॥
yaduttarasyāyanato gataṃ syācchesaṃ tathā dakṣiṇo'yanasya |
tad-ekaṣaṣṭyā dviguṇaṃ vibhaktaṃ sadvādaśaṃ syād-divasapramāṇam ||

According to this, the number of days which have elapsed in the northward course of the Sun (Uttarāyaṇa) or the remaining days in the southward course (Dakṣiṇāyana) doubled and divided by 61, plus 12, is the daytime (in muhūrtas) of the day taken. Hence, the duration of the daytime D_t is given by:

$D_t = \left(12 + \frac{2n}{61}\right)$ muhūrtas, where n denotes the number of days elapsed in Uttarāyaṇa, or the number of days yet to elapse in Dakṣiṇāyana and muhūrta = 1/30th of a civil day (of 24 hours).

The calculation of daytime given in Vedāṅga Jyotiṣa, matches closely with contemporary astronomy calculations. This procedure for calculation of daytime was further improvised by later day Indian astronomers including Āryabhaṭa, Bhāskara and Nīlakaṇṭha Somayājī.

9.5 ĀRYABHAṬĪYA AND THE SIDDHĀNTIC TRADITION

Compared to Vedāṅga Jyotiṣa the later astronomical texts of the siddhāntic period are far more sophisticated and elaborate, with procedures to calculate the planetary positions, eclipses, the times of sunrise and sunset, duration of the day, relation between the shadow of the Sun and the time, and many other quantities of astronomical interest. Āryabhaṭa was the pioneering figure in the tradition of full-fledged mathematical astronomy in India called the siddhāntic tradition, though there were siddhāntas even before Āryabhaṭīya. The major astronomers and their distinctive contributions have been listed in Table 9.2. There were many innovations in techniques and the evolution of ideas in this tradition. Detailed explanations, derivations and demonstrations (geometric and otherwise) are found in the commentaries. This development and refinement continued for over 1200 years.

Āryabhaṭīya is the first extant (discovered and available) text on mathematical astronomy in India. It was a pioneering work that established the framework for mathematical astronomy in India. It contains a systematic treatment of all the traditional astronomical problems. The text was composed in 499 CE. Āryabhaṭīya describes the procedures for calculating the positions of the Sun, the Moon, and the planets, astronomical variables associated with the daily paths of these objects in the sky, especially the Sun, and the important phenomena of eclipses. In all these calculations, trigonometry plays a crucial role. The 'jyārdha', or the half-chord introduced in Āryabhaṭīya is far more convenient than the Greek chord for astronomical computations. This jyārdha is nothing but the modern sine function (apart from a constant factor). A very important recursion relation for the sines and an explicit table of sines are given in Āryabhaṭīya. Āryabhaṭīya has only 121 verses and is organized into 4 parts, namely: Gītikāpāda, Gaṇitapāda, Kālakriyāpāda, and Golapāda. Table 9.5 has a summary of the issues discussed in Āryabhaṭīya.

TABLE 9.5 Organisation of Āryabhaṭīya and the Issues Covered

No.	*Section Name*	*No. of Verses*	*Issues Covered*
1	Gītikāpāda	13 verses	The letter-numeral notation for numbers, concepts of Kalpa and Mahāyuga, revolution numbers of planets and parameters associated with them
2	Gaṇitapāda	33 verses	Square, Square root, cube, cube root, areas of a triangle, a trapezium, and general plane figures, volumes of right pyramids and spheres, value of π, methods for computing Sines geometrically, constructing a sine table, arithmetic progression, summation of first n natural numbers, the sum of their sums, sums of squares and cubes of first n natural numbers, Kuṭṭaka procedure to solve linear indeterminate equations, relative velocities of moving objects, and a problem related to interest calculation
3	Kālakriyāpāda	25 verses	Reckoning of time, calendrical concepts, planetary models (epicycle and eccentric circle theories), and explicit procedures for calculation of planetary positions, etc.
4	Golapāda	50 verses	Problems of spherical astronomy as seen at different latitudes, diurnal problems associated with the motion of the Sun, Moon, and planets on the celestial sphere, situation of the earth and its shape, brightness/darkness of planets, parallax, lunar eclipses, solar eclipses and so on.

According to Āryabhaṭa, the Earth is a sphere at the center of the framework in which stars and planets move. According to a verse in Āryabhaṭīya[18]:

> *"The globe of the Earth stands (supportless) at the center of the circular frame of asterisms surrounded by the orbits (of the planets); it is made up of water, earth, fire, and air and is spherical."*

According to the earlier astronomical works in India and elsewhere, the Earth was stationary, and all the celestial objects rotated in the sky, completing one rotation in one day. The celestial objects rise in the eastern part of the sky, and set in the western part, and are visible when they are above the horizon. However, Āryabhaṭīya presented a different view. The diurnal motion of the celestial objects was mentioned in one of the verses as follows[19]:

corrections specified in later texts). In the case of the ... in India (traditionally only Mercury, Venus, Mars, Jupiter, and Saturn), we obtain the true heliocentric longitude, that is, the longitudes with respect to the Sun after the manda-saṃskāra.

2. *Śīghra-saṃskāra:* This correction is done only for the planets. This converts their heliocentric longitudes (as observed from the Sun) to the geocentric longitudes (as observed from the Earth).

Āryabhaṭīya discussed the above two corrections for the first time in the Indian tradition. In the early 17th century CE, the European astronomer, Kepler had given the correct model for the motion of the planets. According to him, planets move in elliptical orbits around the Sun. The planetary model which is implicit in Āryabhaṭīya and also in the later Indian texts is broadly equivalent to the Kepler model and gives approximately the same final results for the longitudes. There were some problems with the interior planets, Mercury and Venus which were resolved by Nīlakaṇṭha Somayājī in 1500 CE in his Tantra-saṅgraha.

There were some problems associated with the calculational procedure for Mercury and Venus in the traditional Indian model. Around 1500 CE, Nīlakaṇṭha Somayājī modified the procedure based on a detailed scientific analysis and proposed a revised planetary model in his Āryabhaṭīya-bhāṣya and other works. According to this model, the planets move in eccentric orbits around the Sun (that is, the centers of the orbits will be slightly displaced from the Sun), which itself orbits around the Earth. See Figure 9.7 for a simple schematic representation of this model. This was before the well-known heliocentric model of Copernicus in 1542 CE, which has some shortcomings. These are absent in the revised model of Nīlakaṇṭha, which is in a geocentric framework. It is essentially the same as Tycho Brahe's model for the planetary motion proposed around 1580 CE, where the planets move around the Sun, with the Sun itself orbiting the Earth.

पंचांग

9.6 PAÑCĀṄGA – THE INDIAN CALENDAR SYSTEM

Pañcāṅga is the backbone of Indian astrology and calendar. All festivals and local events are based on pañcāṅga, and it plays a vital role in the daily lives of the vast majority of the people in the country. Therefore, there are several regional versions of Pañcāṅga throughout the country to cater to the local requirements. The development of calendric astronomy in India in a systematic manner can be traced back to the Vedāṅga Jyotiṣa, wherein the calendar was formulated based on cycle-periods of five years, called yuga. However, it was the Siddhānta Jyotiṣa, which employed scientific computations that ushered in the refined calendar and Pañcāṅga system that is in vogue today.

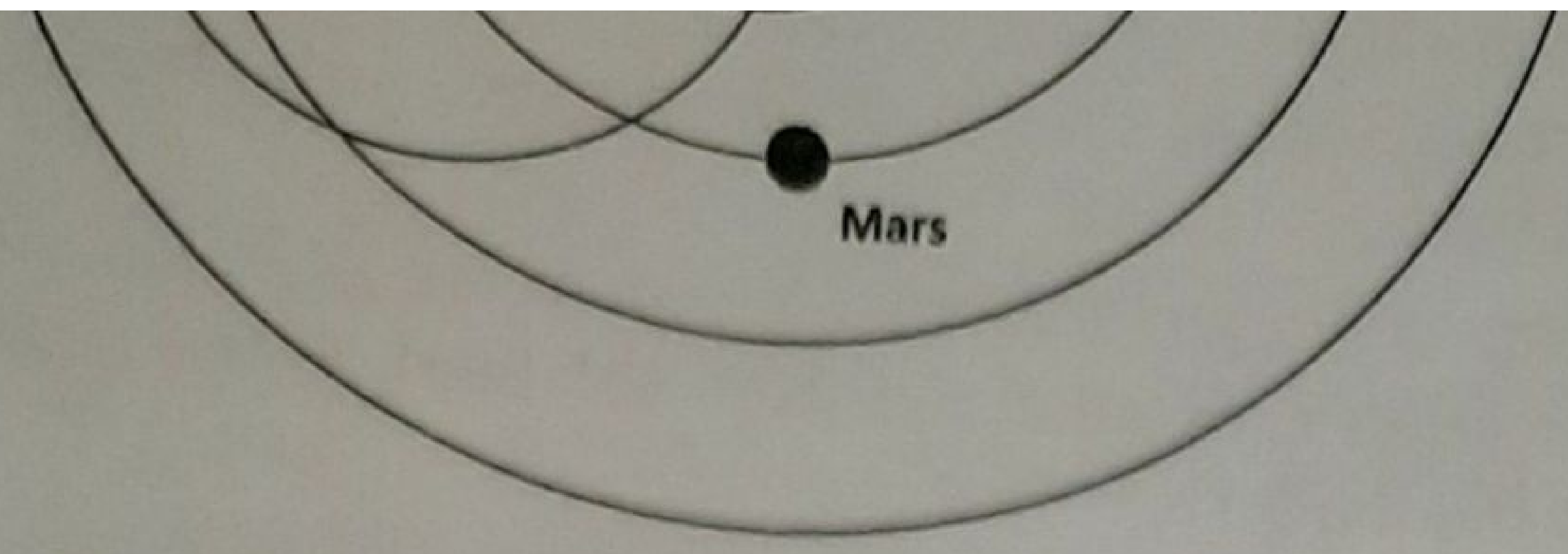

FIGURE 9.7 Nīlakaṇṭha Somayāji's Planetary Model

Many of them employ Sūrya-siddhānta and also use some texts popular in specific regions. In recent times, Graha-lāghava of Gaṇeśa Daivajña and Siddhānta-darpaṇa of Candraśekhara Sāmanta are used to make pañcāṅga calculations. Pañcāṅga, as the name suggests has five components: tithi, nakṣatra, vāra, karaṇa, and yoga. Based on the astronomy concepts developed, pañcāṅga computes the five components at any given instant. In the pañcāṅgas used by the Indian society, such calculations are done, and the results are made available in a ready reckoner format for direct reading and interpretation. The calculations are based on the true longitudes of the Sun and the Moon.

Tithi

Tithi plays a very important role as it determines most of the religious festivals and functions in India. The computational aspects of tithi in a pañcāṅga follows our earlier discussion on tithi. Let θ_M and θ_S be the true Longitudes of the Moon and the Sun in degrees respectively at any instant. Let $\frac{(\theta_M - \theta_S)}{12} = I + f$, where I is an integer, and f is the fractional part. Then I is the number of tithis that have elapsed in the lunar month.

Let us consider two cases for the true longitudes of the Moon (θ_M) and the Sun (θ_S) at any instant. The calculation of tithi is done as follows:

Case 1: (θ_M) = 60° 12′ and (θ_S)= 19° 7′, the computations are as follows:

$$\frac{(\theta_M - \theta_S)}{12} = \frac{(60^\circ\, 12' - 19^\circ\, 7')}{12} = \frac{41^\circ\, 5'}{12} = 3\frac{61}{144}$$

Therefore 3 tithis have elapsed, and the current tithi is 4th, which is Caturthī in Śukla-pakṣa.

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increases by 6°. There are 60 karaṇas in a lunar mo... the current karaṇa is the same as for the number of karaṇas elapsed, and the time elapsed in the current karaṇa is the same as for the tithi except that 12 should be replaced by 6.

To find the karaṇa, we repeat the above calculation with 6 as the divisor.

Case 1:

$$\frac{(\theta_M - \theta_S)}{6} = \frac{(60° 12' - 19° 7')}{6} = \frac{41° 5'}{6} = 6\frac{61}{72}$$

So, 6 karaṇs have elapsed, and the current karaṇā is 7th.

Case 2:

$$\frac{(\theta_M - \theta_S)}{6} = \frac{(561° 2' - 219° 17')}{6} = \frac{341° 45'}{6} = 56\frac{23}{24}$$

So, 56 karaṇs have elapsed, and the current karaṇa is 57th.

Nakṣatra

As we have already seen, the ecliptic is divided into 27 equal parts called nakṣatras beginning with Aśvinī and ending with Revatī. Hence each nakṣatra corresponds to $\frac{360 \times 60}{27} = 800$ minutes along the ecliptic. The nakṣatra at any instant refers to the particular portion of the ecliptic in which the Moon is situated. When the longitude of the Moon in minutes is divided by 800 the quotient gives the number of nakṣatras that have elapsed, and the remainder corresponds to the minutes covered by the Moon in the present nakṣatra.

If we consider the above example, the nakṣatras for the two longitudes of the Moon will be as follows:

To find the nakṣatra, we first convert θ_M into minutes and divide it by 800.

Case 1: (θ_M) = 60° 12′ =3612′.

Nakṣatra = $\frac{3,612}{800} = 4\frac{103}{200}$. Therefore 4 nakṣatras have elapsed, and the current nakṣatra is 5th (Mṛgaśiras).

Case 2: (θ_M) = 201° 2′ = 12,062′.

Nakṣatra = $\frac{12,062}{800} = 15\frac{31}{400}$. Therefore 15 nakṣatras have elapsed, and the current nakṣatra is 16th (Viśākhā).

9.7 ASTRONOMICAL INSTRUMENTS (YANTRAS)

Astronomy, unlike certain other disciplines like architecture and metallurgy, is more a science of observation and computation. The positions and movements of heavenly bodies have to be observed and recorded very accurately before a theory to explain their motions can be propounded. The theories have to be revised if their predictions do not tally with the observations. Also, for several astronomical computations, the time since the sunrise has to be measured precisely. In the case of motions of stars, planets and other luminaries in the sky, visual observations are not likely to be accurate. Therefore, it is necessary to devise instruments to ascertain the positions and motions of heavenly bodies and measure the duration of time. Ancient Indians recognized this and developed a variety of instruments, (yantras). The instruments that were used in Indian astronomy include the water clock (ghaṭī-yantra), gnomon (śaṅku), cross-staff (yaṣṭi-yantra), armillary sphere (gola-yantra), board for the sun's altitude (phalaka-yantra), sundial (kapāla-yantra) to name a few. We shall see a brief description of some of these.

Gnomon

The simplest instrument is just a straight, vertical stick (śanku) which has a pointed tip. This is called a 'gnomon' in astronomy, which along with a rope formed a basic but very powerful instrument in Indian astronomy. The Indian texts describe methods to find various astronomical quantities like the directions, time of the day, the latitude of the place, and the longitude of the Sun, etc. from the shadow of the gnomon. When the shadow of the pillar was at its shortest length on any day, the gnomon indicated midday. To obtain some idea of the time of the morning or afternoon, an observer can assess the length and direction of the shadow. The first use of the gnomon in India appears to be to fix the cardinal points.

Procedure for Fixing the Cardinal Direction

One of the first requirements is to fix the cardinal directions (east, west, south, north) in a given location. Kātyāyana-śulba-sūtra gives the following method to draw the east-west line. Figure 9.8 depicts this procedure pictorially. Place a gnomon OX, exactly vertically on level ground, so that its tip at X points at the zenith Z. Draw a circle with a suitable radius with the base of the gnomon O as the Centre. Mark the points W' and E' where the tips of the shadow of the gnomon are on the circle, in the forenoon and the afternoon respectively. Then, $E'W'$ is in the east-west direction. This is because the shadows of the same length would be

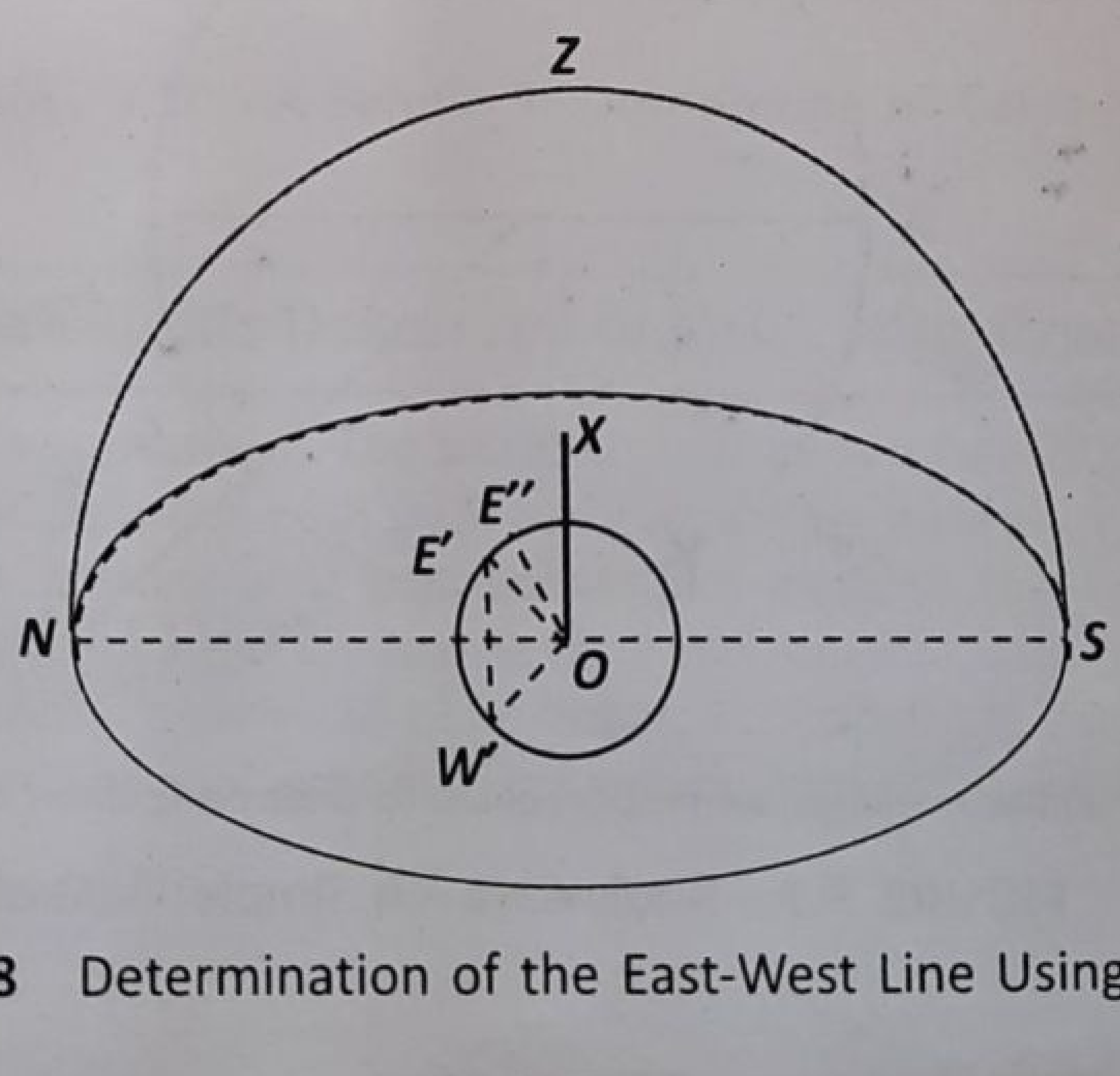


FIGURE 9.8 Determination of the East-West Line Using a Gnomon

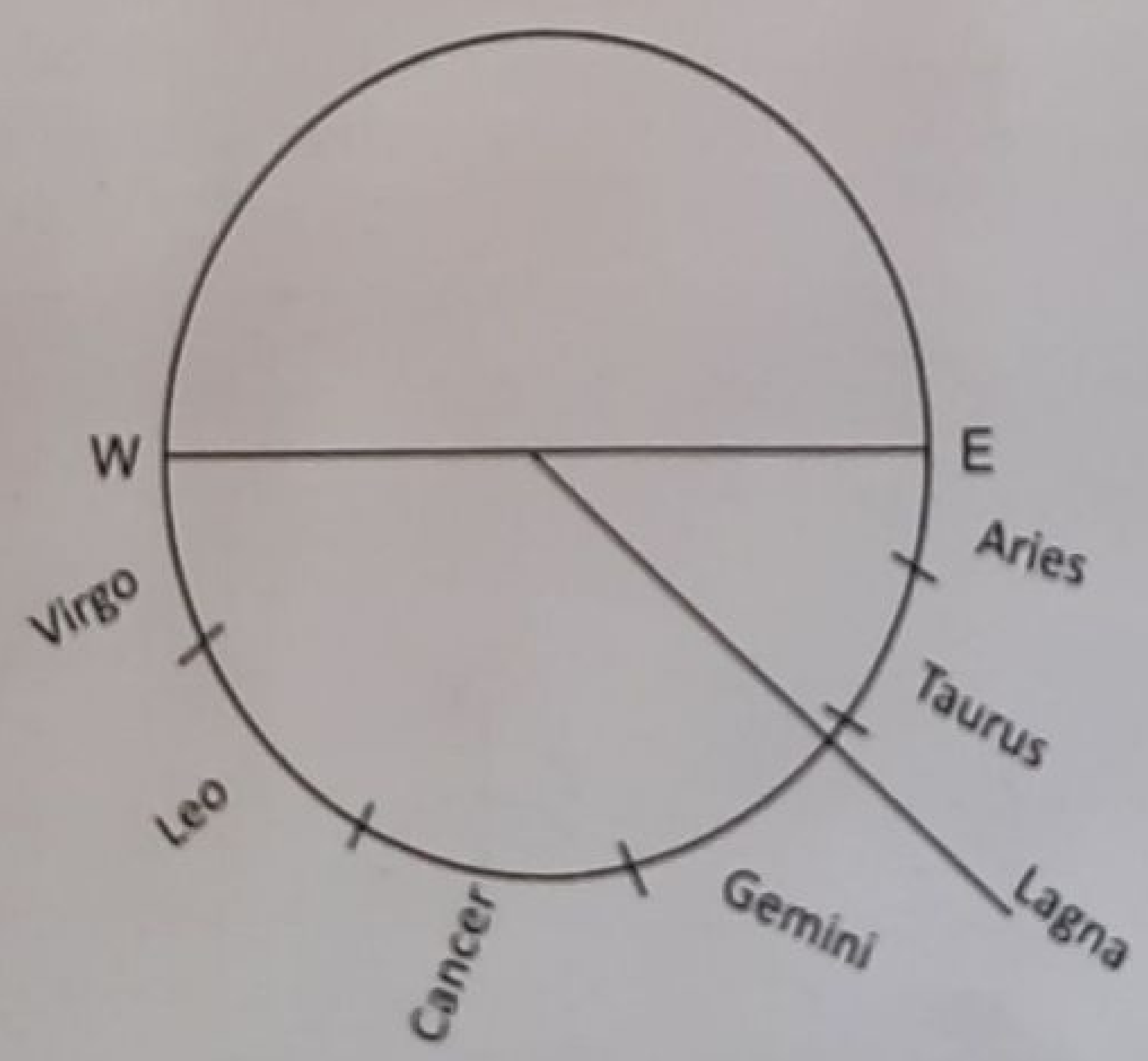
symmetrically situated with respect to the north-south line, and $E'W'$ is perpendicular to that and is the east-west line. One can draw a line parallel to this passing through O, for locating the east and west points at O. There is a slight inaccuracy in the above procedure, due to the change in the declination of the Sun during the day. So, if W' and E'' are the points at which the shadows touch the circle, the east-west line is $E'W'$, where E' is slightly displaced from E''. The expression for the amount of displacement is discussed in a commentary on Brāhmasphuṭa-siddhānta of Brahmagupta and many texts which followed it.

Armillary Sphere

In its simplest form, an armillary sphere is a ring of bronze fixed in the plane of the equator. When the shadow of the upper half exactly covered the lower half, it indicates an equinox. However, the armillary sphere described by Bhāskara II and other Indian astronomers is elaborate. It consisted of three spheres; a bhagola or the sphere of the fixed stars, a khagola or the sphere of the sky outside the bhagola, and a third sphere known as dṛggola outside the khagola, in which the circles forming both the spheres khagola and bhagola are mixed. This can be used to find the time since sunrise and the 'lagna' at that time[20].

Nāḍīvalaya-yantra

The Nāḍīvalaya is a large circular wooden disc with an axis in the centre (see Figure 9.9). It is divided into 60 ghaṭikās and also into 12 signs of the ecliptic with variable arcs corresponding to the periods of their risings in the place of observation. The twelve periods thus marked are further subdivided into finer units. The signs are inscribed in the reverse order of the signs, that is first Aries, then Taurus to the West of Aries, and so on. It is placed in a plane parallel to the equinoctial so that the axis in the centre points towards the north pole. The longitude of the Sun in signs, and degrees for the sunrise of the given day is obtained by calculation and that position is marked on the circle. The disc is rotated around the axis so that the shadow of the axis falls on the mark made for the position of the Sun at sunrise. Using this method, the disc is fixed in position. As the sun rises, the shadow of the axis advances from the marks made for the point of the sunrise to the nadir. The number of ghaṭis will be seen between the point of sunrise to the position of the shadow which will also indicate the lagna or the sign on the eastern horizon.



* The spacing of zodiac markings are proportionate to their rising times at the place installation.

FIGURE 9.9 Nāḍīvalaya – A Simple Illustration

Gola-yantra

The armillary sphere and the Nāḍīvalaya described above will measure the time and indicate only the lagna after sunrise on a day when the sky is clear. However, it is necessary to measure time during the night also and to determine the lagna at a particular instant. For this purpose, Indian astronomers had devised other instruments. One of these was Gola-Yantra, which is made of wood, perfectly spherical, and uniformly dense all around but light (in weight). According to Āryabhaṭa, it should be made to rotate keeping with time with the help of mercury, oil, and water and by proper application of astronomical principles. Sūrydeva a commentator of this work provides details of how to construct this instrument.

Cakra-yantra

Cakra-Yantra is made in the form of a plate of metal or seasoned wood and a needle is fixed at the center. When the instrument is illuminated by the rays of the Sun on both sides, the shadow of the needle will give the angular height of the Sun. For his observatories at Jaipur and Varanasi, Raja Sawai Jai Sıngh built Cakra Yantras which are very large instruments. There are two instruments at Jaipur each 6 feet in diameter and one at Varanasi, 3 feet 7 inches in diameter, one inch thick and two inches broad, faced with brass, on which degrees and minutes are marked They are mounted on pillars and fixed so as to revolve around an axis parallel to the earth's axis. The axis carries a pointer, which indicates the hour angle on the fixed circle. Figure 9.10 is a simple representation of a Cakra-Yantra.

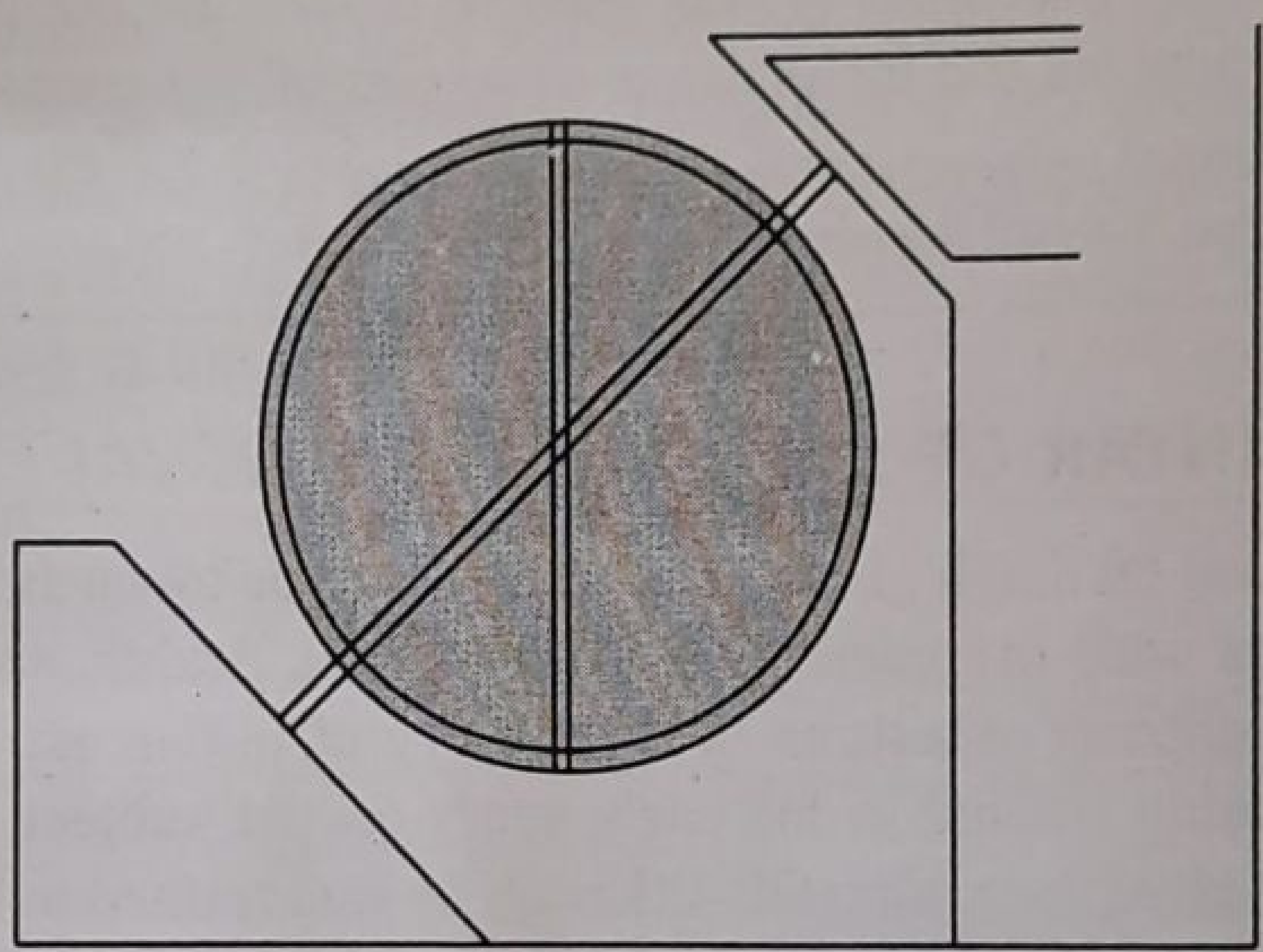


FIGURE 9.10 A Simple Representation of Cakra-Yantra

Astronomical Instruments Described in Siddhānta-śiromaṇi

Astronomy is an observational science. The positions and movements of heavenly bodies need to be continuously observed and recorded with good accuracy before a theory to explain their motions can be propounded. Therefore, astronomical instruments play a crucial role in advancement of theory in astronomy. Indian astronomers have devised several instruments over time. All these instruments constituted astronomical observatories in older times. In Siddhānta-śiromaṇi of Bhāskarācārya, we find a detailed description of several astronomical instruments. Some of them with brief descriptions are given below:

1. **Gola-yantra, the armillary sphere:** A sphere in which all movable and fixed circles are designed. An observer can perform observations sitting within the sphere itself. This instrument serves the purpose of an astrolabe. Bhāskarācārya performed lunar observations with the help of this instrument and compiled his Bījopanaya.
2. **Cakra-yantra:** A wooden or metallic wheel-like structure with an axis fixed in a hole at its centre. It was used to determine longitudes and latitudes of planets.
3. **Cāpa-yantra:** This is half the structure of Cakra yantra.
4. **Turīya-yantra:** This constitutes one quadrant of the Cakra yantra with a stick or nālika (tube) for observing celestial bodies in order to determine their zenith distances and altitudes.
5. **Nāḍīvalaya:** A cakra in the plane of the equator used to determine directly timings of rising and setting of signs.
6. **Ghatī-yantra:** A droṇa (bowl-shaped) water-clock with a hole of standardized size at its bottom.
7. **Nara or Śaṅku:** (the gnomon) made of Ivory or a metal.
8. **Phalaka-yantra:** A plank with a circle of radius 30 aṅgulas drawn on it. The circle graduated in ghaṭīs and degrees. When hanging in the plane of vertical circle, the instrument can read zenith distance directly through graduations on the circle.
9. **Dhī-yantra:** It is a simple stick instrument (augmented by a plumbline-like device to assign vertical direction). It was used to determine the heights and distances of objects by measuring the inclinations or angles of elevation at different points.

Source: **Sharma, S.D. (2000).** "Astronomical Observatories", Chapter 11 in, Sen, S.N and Shukla, K.S. (Eds.), *History of Astronomy in India*, Indian National Science Academy, 2nd ed., pp. 367–394.

9.8 JANTAR MANTAR OF RĀJĀ JAI SINGH SAWAI

development

The efforts of Raja Sawai Jai Singh (1686–1743 CE) of Jaipur towards the fostering of scientific observational practices with the combined use of Indian, Arabic, and European advances in astronomy mark a significant milestone in the history of Indian astronomy. Jai Singh was an astronomer himself, being trained in his early years on the subject. He collected texts of all three traditions and studied them himself. Although he was trained in Indian astronomy he was not satisfied with the use of old data on astronomical constants. Therefore, he prepared metallic instruments after making some improvements but found that the accuracy of his instruments was still not satisfactory. He felt the need for making gigantic instruments that are sturdy, free from wear and tear, and not affected by changing weather conditions, etc. Further, Jai Singh believed that European instruments were not large and suffered from errors. Therefore, he took a different approach to build astronomical instruments and created a versatile astronomical observatory by setting up a group of essential instruments in one place. These observatories known as Jantar Mantar were built in five places: Delhi, Jaipur, Varanasi, Ujjain, and Mathura during his regime. The observatory at Mathura was located on the banks of Yamuna in a fort built by Raja Man Singh of Amber. Due to multiple attacks, there are not even debris of the observatory anymore. The last remains of the Mathura Jantar Mantra were demolished in 1857 CE when a contractor, Jyoti Prasad bought the fort.

When Jai Singh constructed the Delhi observatory, he ensured that the rules of geometry are properly adhered to, for adjustments to meridians and latitudes of the place so that all errors were rectified. The Delhi observatory was completed by 1724 CE. He erected instruments of his own invention, such as Jai Prakāśa-yantra, Rāma-yantra, and Samrāṭ-yantra. Samrāṭ-yantra had semidiameters of 18 cubits (1 cubit = 20 inches) with a thin layer of lime and stone which provided good stability to the instruments. The observations obtained from the instruments erected at the Delhi Jantar Mantar were found to give measurements that tallied well with other known results. To confirm the reliability of the results obtained through observations, he constructed similar observatories in Jaipur, Ujjain, Varanasi, and Mathura.

The Jantar Mantar deployed a variety of known astronomical instruments such as Cakra Yantra, Nāḍīvalaya, and Kapāla-yantra. When Jai Singh constructed the Jantar Mantar, he utilized the tradition of astronomical techniques in India but also made some innovations by bringing in the astronomical techniques he learnt from the Arabic and European astronomers. Let us see some details about Samrāṭ-yantra.

Yantra Samrāṭ is a big instrument with a gnomon wall which is set in the precisely determined north-south direction and is made up of lime brick stones. This is 3 hastas (1 hasta = 18 inches) in thickness and 24 hastas in length. Its upper part is slanting, made up of stones, and points towards Polaris. In the south the root side is 4.5 hastas in height and on the front side in the north it is three inches short of 15 hastas. It has ladders in between. On the eastern and western sides, there are two stony arcs with radii 6 hastas and are more than a quadrant each in size. These have a breadth of about 4 hastas and a thickness of 9.5 aṅgulas. Their centres lie on the śankupālis (the edges of the gnomon wall). On each arc, there are quadrants which on edges have divisions in 15 equal ghatis. Each division is further graduated in uniform six subdivisions which are a little less than an aṅgula in measure. On the śaṅkupālis (on both edges of the gnomon wall) there are two small iron rings, whose centers coincide with the centers of the cāpa-pālis (the edges of the arcs). Figure 9.11 is a simple sketch of the instrument.

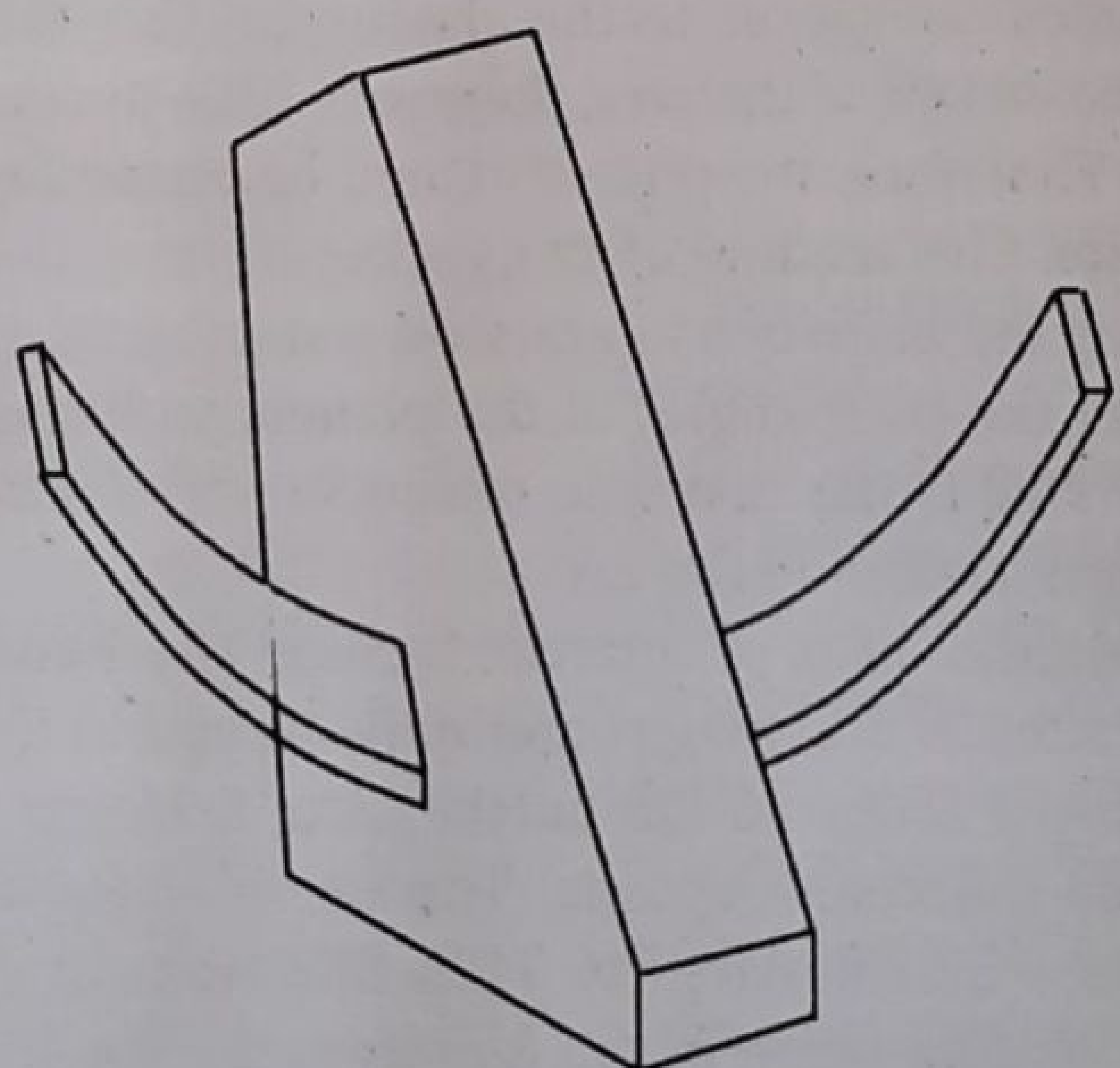


FIGURE 9.11 A Simple Sketch of Samrāṭ-yantra

FIGURE 9.12 East Side of Samrāṭ-yantra 2 at Delhi Jantar Mantar
Source: https://commons.wikimedia.org/

Using this instrument, time can be read in degrees from the top of the gnomon up to the shadow on the cāpa-pāli (in units of ghaṭī) during the daytime. If the shadow of the gnomon falls on the western side, then it indicates the remaining time for midday. On the contrary, if the shadow falls on the eastern side it indicates the time past midday. Steps built by the side of arcs enables one to reach out closer to the shadow so that the measurement is accurate. However, on account of the weight of the arcs, they are a little inclined from the actual positions near the terminal edges. Therefore, the time obtained by observing the shadows at the edges is likely to be less accurate. The shadow of the śaṅkupālī from the Moon is not so clear as it is from the Sun. Moreover, the planets and stars do not produce shadow at all. Nevertheless, it is possible to determine the hour angles of the planets and those of stars with the help of this instrument[21]. Figure 9.12 is the east side of the Samrāt Yantra 2 in Delhi Jantar Mantar, where we can see the stairs alongside the arc.

Other archaeoastronomical sites of interest include the Neolithic site of Burzahom (in Kashmir), and megalithic sites in Brahmagiri and Hanamsagar in Karnataka in the South India. The Burzahom, site is located about 10 km northeast of Srinagar in the Kashmir Valley on a terrace of Late Pleistocene-Holocene deposits. The site is dated to around 3000–1500 BCE. A stone slab of size 48 cm × 27 cm, dated to 2125 BCE was obtained that shows two bright objects in the sky with a hunting scene in the foreground.

The megalithic stone circles of Brahmagiri in the Chitradurga district of Karnataka in South India, have been dated to 900 BCE. Studies pertaining to this suggest that site lines from the center of a circle to an outer tangent of another circle point to the directions of the sunrise and full moon rise at the time of the solar and lunar solstices and equinox. Hanamsagar in Karnataka is another megalithic site with stone alignments pointing to cardinal directions. There are 2,500 stones arranged in a square of side 600 meters with 50 rows and 50 columns with a separation between stones of about 12 m. The lines are oriented in cardinal directions. It has been argued that the directions of summer and winter solstice can be fixed in relation to the outer and the inner squares. Other astronomical observations such as estimating the time of the day using the shadow, the prediction of months, seasons, and passage of the year could have been possible using the structure in the site[22].

SUMMARY

- Knowledge of astronomy was widely used by all sections of Indian society, not just by the subject matter experts.
- Good knowledge of astronomical concepts of ancient Hindus is also evident from several Vedic texts.
- Āryabhaṭīya is the first text on astronomy in India. It established the framework for mathematical astronomy in India. It contains a systematic treatment of all the traditional astronomical problems.
- The elapsed time for the Sun to the same star, which is nothing but the period of the revolution of the Earth around the Sun, is the solar year.
- It had been noticed that the Sun and the Moon return together to nearly the same position in the framework of stars, after five years. The concept of Yuga was introduced to synchronize the Solar and Lunar calendars.
- A lunar month (cāndra-māsa) is the time interval between two new Moons (Amāvāsyā), or two full Moons (Pūrṇimā). It consists of two pakṣas; Śukla-pakṣa (bright fortnight) from Amāvāsyā to Pūrṇimā, and the Kṛṣṇa-pakṣa (dark fortnight) from Pūrṇimā to Amāvāsyā.
- In the Indian system, the ecliptic is divided into 27 equal divisions, known as nakṣatras. Each day would be associated with a nakṣatra.
- Tithi captures the angular separation between the Sun and the Moon. It is a time-unit during which the angle between the Sun and the Moon (as viewed from the Earth) increases precisely by 12°.
- The Vedāṅga Jyotiṣa gives rules for calculating some astronomical quantities related to the Sun and the Moon. The astronomical texts of the siddhānta period are far more sophisticated and elaborate, with procedures to calculate planetary positions, eclipses, etc.
- Around 1500 CE, Nīlakaṇṭha Somayājī modified the procedure based on a detailed scientific analysis and proposed a revised planetary model in his Āryabhaṭīya-bhāṣya and other works.
- There are several regional versions of Pañcāṅga throughout the country to cater to the local requirements. They all employ Sūrya-siddhānta to compute the Pañcāṅga. In recent times, Graha-lāghava of Gaṇeśa Daivajña is used to make pañcāṅga calculations.
- The instruments used in Indian astronomy include the water clock (ghaṭi-yantra), gnomon (śaṅku), cross-staff (yaṣṭi yantra), armillary sphere (gola-yantra), board for the sun's altitude (phalaka-yantra), and sundial (kapāla-yantra).
- The simplest astronomical instrument is just a straight, vertical stick (śaṅku) which has a pointed tip. This is called a 'gnomon' in astronomy.
- In its simplest form, an armillary sphere is a ring of bronze fixed in the plane of the equator.
- The efforts of Raja Sawai Jai Singh of Jaipur towards the fostering of scientific observational practices with the combined use of Indian, Arabic, and European advances in astronomy mark a significant milestone in the history of Indian astronomy.